Kenya ENSO Explorer: Citable Scientific Methods Monograph & Companion Documentation

Theoretical Foundations, Empirical Baselines, and Remote Sensing Analytics for Subnational Climate Risk Assessment

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1 Overview & Citation Guide

2 Executive Summary

This scientific monograph provides formal methodological documentation, empirical baselines, statistical derivations, and data provenance for the **Kenya ENSO Explorer** (https://africaadaptationatlas.cgiar.org/).

The platform provides subnational decision-support across Kenya's 47 counties to evaluate how equatorial ocean dynamics (El Niño–Southern Oscillation, Indian Ocean Dipole, and Western Pacific anomalies) govern precipitation regimes, rangeland pasture biomass, physical flood inundation, and agricultural capital at risk.

2.1 Suggested Citation

> Steward, P., African Agriculture Adaptation Atlas Science Team, & Kenya Meteorological Department (2026). *Kenya ENSO Explorer: Citable Scientific Methods Monograph & Companion Documentation* (Version 3.11.0). CGIAR Climate Action / Alliance of Bioversity International and CIAT. https://doi.org/10.5281/zenodo.10849201

2.1.1 BibTeX Entry

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  author = {Steward, Pete and {African Agriculture Adaptation Atlas Science Team} and {Ke
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```

2.2 Structure of this Monograph

- **Chapter 1: County Context, Demography & Social Inclusion**: Demographic baselines, population densities, subcounty area geometry, and gender equity indicators.

- **Chapter 2: Equatorial Ocean Teleconnections & Seasonal Forecasting**: Sea surface temperature indices (RONI, ONI, DMI, Western-V), forecast plume verification, and ranked probability skill scores (RPSS).
- **Chapter 3: Precipitation Climatology, Empirical Quantiles & Regimes**: Mathematical rationale for non-parametric quantiles, positive rainfall skewness, and independent seasonal sequence classification.
- **Chapter 4: Satellite Earth Observation, Hazards & Exposure**: Sentinel-1 SAR flood extent (Copernicus GFM), JRC return period hydrodynamics, and MODIS NDVI pasture biomass anomalies.
- **Chapter 5: Agricultural Production & Economic Valuation**: KNBS agricultural survey detrending, subcounty capital weighting, and shoat-to-maize terms of trade.
- **Chapter 6: Master Dataset Catalogue & Provenance**: Full metadata contracts, producers, and licensing for all 23 platform datasets.
- **Chapter 7: Controlled Vocabulary & Glossary**: Standardized definitions, equations, and institutional references for 65 domain terms.

3 County Context, Demography & Social Inclusion

4 County Context, Demography & Social Inclusion

This chapter details the demographic, administrative, and gender vulnerability baselines across Kenya's 47 counties and sub-counties. It documents spatial geometric intersections over UN OCHA COD-AB boundaries, population density models from the KNBS 2019 National Census, and the multidimensional Gender Equality and Social Inclusion (GESI) indicators.

4.1 Figure 1.3: Socio-Economic & Gender Vulnerability Profile

Section in Platform: Section 1.3 | **Target Identifier:** `fig_1_3` | **Output Type:** Chart

> **Policy / Scientific Question:** *How does the county compare against the national benchmark across key social and economic vulnerability indicators?*

4.1.1 Scientific Purpose & Context

Quantify and communicate Socio-Economic & Gender Vulnerability Profile across Kenyan counties to inform anticipatory action and climate adaptation planning.

4.1.2 Mathematical Formulation & Algorithmic Approach

Baseline Reference Period: 1991–2020 WMO standard normal

Gender Equality and Social Inclusion (GESI) indicators extracted from KNBS County Gender Data Sheets (Oct 2025 edition). Evaluates female headship, agricultural land ownership titles, agricultural labor share, and clean cooking fuel access relative to national medians.

Applies non-parametric or standardized metrics aligned with Decision D43 and D53 to avoid distributional distortion.

Governing Mathematical Formulations:

$$\Delta_{\text{GESI}} = I_c - \text{Median}_{\text{Kenya}}$$

4.1.3 Underlying Data Assets & Inputs

Provenance Key	Dataset Title	Spatial Resolution	Temporal Span
`kenya-county-gender-datasheets`	Kenya County Gender Data Sheets, 2025	kenya	None

Served Parquet Files in Platform: `gesi.parquet`, `gesi_v2.parquet`

4.1.4 Methodological Assumptions & Analytical Limitations

- County aggregations rely on official polygon boundaries from UN OCHA COD-AB standard.
- Sub-seasonal temporal variations within rolling seasons are smoothed over seasonal totals.
- Data latency reflects standard publication cadences of underlying satellite and administrative surveys.

4.1.5 Controlled Vocabulary Keywords

`GESI` • `Female Headship` • `Land Tenure` • `KNBS Gender Sheets`

4.2 Table 1.1: Sub-county Baseline Geography & Rainfall Climatology

Section in Platform: Section 1.0 | **Target Identifier:** `table_1_0` | **Output Type:** Table

> **Policy / Scientific Question:** *What are the baseline geographic dimensions, population, and rainfall patterns across sub-counties?*

4.2.1 Scientific Purpose & Context

Quantify and communicate Sub-county Baseline Geography & Rainfall Climatology across Kenyan counties to inform anticipatory action and climate adaptation planning.

4.2.2 Mathematical Formulation & Algorithmic Approach

Baseline Reference Period: 1991–2020 WMO standard normal

Spatial polygon intersection over UN OCHA COD-AB boundaries; population density = headcount / area (km²); 1981–2025 CHIRPS v3 monthly accumulation aggregated to subcounty administrative boundaries.

Applies non-parametric or standardized metrics aligned with Decision D43 and D53 to avoid distributional distortion.

Governing Mathematical Formulations:

$$\rho_{sc} = \frac{\text{Pop}_{sc}}{\text{Area}_{sc}}$$

$$\bar{R}_{s,sc} = \frac{1}{N} \sum_{y=1}^{N} \sum_{m \in s} R_{y,m,sc}$$

4.2.3 Underlying Data Assets & Inputs

Provenance Key	Dataset Title	Spatial Resolution	Temporal Span
`kenya-population-knbs-census`	KNBS 2019 Census Population Counts by Co	kenya — counties (47 admin-1 units); a sub-county table is included in the KNBS universe	2019
`africa-precipitation-monthly-seasonal`	Monthly and Seasonal Precipitation from	africa — 0.05 deg (~5.5 km, CHIRPS native)	1981-01 to present (open-ended)
`kenya-admin-codab`	Kenya Administrative Boundaries, Countie	kenya — Admin 1 (47 counties) and admin 2 (290 sub-counties)	2019-10 to 2025-01

Served Parquet Files in Platform: `population_knbs_census_adm1.parquet`, `subcounty_rainfall_climatology.parquet`

4.2.4 Methodological Assumptions & Analytical Limitations

- County aggregations rely on official polygon boundaries from UN OCHA COD-AB standard.
- Sub-seasonal temporal variations within rolling seasons are smoothed over seasonal totals.

- Data latency reflects standard publication cadences of underlying satellite and administrative surveys.

4.2.5 Controlled Vocabulary Keywords

`KNBS Census` • `COD-AB` • `MAM` • `OND` • `Bimodal Normal`

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5 Equatorial Ocean Teleconnections & Seasonal Forecasting

6 Equatorial Ocean Teleconnections & Seasonal Forecasting

This chapter presents the physical oceanographic indices, dynamical model plumes, and statistical verification frameworks governing Kenyan seasonal climate predictability. It establishes the detrended Relative Oceanic Niño Index (RONI), Indian Ocean Dipole Mode Index (DMI), multi-model forecast plume ensembles (SINTEX-F, IRI), and Ranked Probability Skill Score (RPSS) cross-validation protocols.

6.1 Advisory Card 2.0: Kenya Meteorological Department (KMD) / KMSA ClimWeb CAP Weather Advisories

Section in Platform: Section 2 | **Target Identifier:** `box_2_0` | **Output Type:** Table

> **Policy / Scientific Question:** *What active severe weather alerts, heavy rainfall warnings, and precautionary advisories are currently in effect for this county from the Kenya Meteorological Department?*

6.1.1 Scientific Purpose & Context

Quantify and communicate Kenya Meteorological Department (KMD) / KMSA ClimWeb CAP Weather Advisories across Kenyan counties to inform anticipatory action and climate adaptation planning.

6.1.2 Mathematical Formulation & Algorithmic Approach

Baseline Reference Period: 1991–2020 WMO standard normal

Official Kenya Meteorological Department (KMD) seasonal climate advisory synthesis and sectoral contingency protocols (Decision D40 canonical naming).

Applies non-parametric or standardized metrics aligned with Decision D43 and D53 to avoid distributional distortion.

6.1.3 Underlying Data Assets & Inputs

Provenance Key	Dataset Title	Spatial Resolution	Temporal Span
`kmd-official-advisories`	kmd-official-advisories	County (ADM1)	Historical multi-decadal

Served Parquet Files in Platform: `kmd_cap_county_active.parquet`

6.1.4 Methodological Assumptions & Analytical Limitations

- County aggregations rely on official polygon boundaries from UN OCHA COD-AB standard.
- Sub-seasonal temporal variations within rolling seasons are smoothed over seasonal totals.
- Data latency reflects standard publication cadences of underlying satellite and administrative surveys.

6.1.5 Controlled Vocabulary Keywords

`KMD` • `Canonical Naming` • `Sectoral Advisory` • `Common Alerting Protocol`

6.2 Figure 2.1: Empirical Tercile Probability Distribution & Forecast Plume (with 45-Year LOOCV Hindcast Skill)

Section in Platform: Section 2 | **Target Identifier:** `fig_4_1` | **Output Type:** Chart

> **Policy / Scientific Question:** *What is the empirical probability of a wetter, near-normal, or drier season based on nearest historical ocean analogues, what is the retrospective hindcast skill (RPSS & accuracy, 1981–2025), and how is the Pacific ENSO state projected to evolve?*

6.2.1 Scientific Purpose & Context

Quantify and communicate Empirical Tercile Probability Distribution & Forecast Plume (with 45-Year LOOCV Hindcast Skill) across Kenyan counties to inform anticipatory action and climate adaptation planning.

6.2.2 Mathematical Formulation & Algorithmic Approach

Baseline Reference Period: 1991–2020 WMO standard normal

Forecast seasonal rainfall tercile probabilities conditioned on equatorial ocean state distributions across Pacific (RONI/ONI) and Indian (DMI) basins.

Applies non-parametric or standardized metrics aligned with Decision D43 and D53 to avoid distributional distortion.

Governing Mathematical Formulations:

$$P(\text{Tercile}_k \mid \Phi_{\text{Ocean}}) = \frac{n(\text{Tercile}_k \cap \Phi_{\text{Ocean}})}{n(\Phi_{\text{Ocean}})}$$

6.2.3 Underlying Data Assets & Inputs

Provenance Key	Dataset Title	Spatial Resolution	Temporal Span
`enso-driver-indices`	Ocean Climate Teleconnection Drivers (Pa	global-oceans, equatorial-pacific, indian-ocean	1950-01 to present (open-ended)
`africa-precipitation-monthly-seasonal`	Monthly and Seasonal Precipitation from	africa — 0.05 deg (~5.5 km, CHIRPS native)	1981-01 to present (open-ended)

Served Parquet Files in Platform: `driver_indices.parquet`, `enso_hindcast_skill.parquet`

6.2.4 Methodological Assumptions & Analytical Limitations

- County aggregations rely on official polygon boundaries from UN OCHA COD-AB standard.
- Sub-seasonal temporal variations within rolling seasons are smoothed over seasonal totals.
- Data latency reflects standard publication cadences of underlying satellite and administrative surveys.

6.2.5 Controlled Vocabulary Keywords

`RONI • DMI • Tercile Probabilities • Conditional Probability`

6.3 Figure 2.2: Historical Analogue Scenario Multi-Hazard Diagnostics

Section in Platform: Section 2 | **Target Identifier:** `fig_4_2` | **Output Type:** Chart

> **Policy / Scientific Question:** *What multi-hazard impacts occurred when this specific analogue ocean condition was observed in the past, and how did compound climate shocks propagate through crop production, rangeland pasture, and market purchasing power?*

6.3.1 Scientific Purpose & Context

Quantify and communicate Historical Analogue Scenario Multi-Hazard Diagnostics across Kenyan counties to inform anticipatory action and climate adaptation planning.

6.3.2 Mathematical Formulation & Algorithmic Approach

Baseline Reference Period: 1991–2020 WMO standard normal

Historical multi-driver analog identification matching joint Pacific RONI/ONI and Indian Ocean DMI phases to past seasons with multi-hazard impacts.

Applies non-parametric or standardized metrics aligned with Decision D43 and D53 to avoid distributional distortion.

Governing Mathematical Formulations:

$$\text{Distance}(y) = \sqrt{w_1(z_{\text{RONI},y} - z_{\text{target}})^2 + w_2(z_{\text{DMI},y} - z_{\text{target}})^2}$$

6.3.3 Underlying Data Assets & Inputs

Provenance Key	Dataset Title	Spatial Resolution	Temporal Span
`enso-driver-indices`	Ocean Climate Teleconnection Drivers (Pa	global-oceans, equatorial-pacific, indian-ocean	1950-01 to present (open-ended)
`africa-precipitation-monthly-seasonal`	Monthly and Seasonal Precipitation from	africa — 0.05 deg (~5.5 km, CHIRPS native)	1981-01 to present (open-ended)

Served Parquet Files in Platform: `driver_indices.parquet`, `chirps_county_monthly.parquet`

6.3.4 Methodological Assumptions & Analytical Limitations

- County aggregations rely on official polygon boundaries from UN OCHA COD-AB standard.
- Sub-seasonal temporal variations within rolling seasons are smoothed over seasonal totals.
- Data latency reflects standard publication cadences of underlying satellite and administrative surveys.

6.3.5 Controlled Vocabulary Keywords

`Analog Year` • `Joint Ocean Phase` • `RONI` • `DMI`

6.4 Figure 5.1: Physical Teleconnection Mechanics: Pacific-to-Kenya Teleconnection

Section in Platform: Section 5.1 | **Target Identifier:** `fig_5_1` | **Output Type:** Diagram

> **Policy / Scientific Question:** *What physical atmospheric wave dynamics connect equatorial Pacific warming to East African rainfall?*

6.4.1 Scientific Purpose & Context

Quantify and communicate Physical Teleconnection Mechanics: Pacific-to-Kenya Teleconnection across Kenyan counties to inform anticipatory action and climate adaptation planning.

6.4.2 Mathematical Formulation & Algorithmic Approach

Baseline Reference Period: 1991–2020 WMO standard normal

Physical teleconnection mechanics: Walker circulation displacement, Indian Ocean dipole zonal wind reversals, and upper-level divergence modulating East African convective rainfall.

Applies non-parametric or standardized metrics aligned with Decision D43 and D53 to avoid distributional distortion.

6.4.3 Underlying Data Assets & Inputs

Provenance Key	Dataset Title	Spatial Resolution	Temporal Span
`enso-driver-indices`	Ocean Climate Teleconnection Drivers (Pa	global-oceans, equatorial-pacific, indian-ocean	1950-01 to present (open-ended)

Served Parquet Files in Platform: `driver_indices.parquet`

6.4.4 Methodological Assumptions & Analytical Limitations

- County aggregations rely on official polygon boundaries from UN OCHA COD-AB standard.
- Sub-seasonal temporal variations within rolling seasons are smoothed over seasonal totals.
- Data latency reflects standard publication cadences of underlying satellite and administrative surveys.

6.4.5 Controlled Vocabulary Keywords

`Walker Circulation` • `Convective Divergence` • `Physical Teleconnection`

6.5 Table 2.1: Two-Season Compound Agricultural & Livelihood Outcome Matrix

Section in Platform: Section 2 | **Target Identifier:** `table_2_1` | **Output Type:** Table

> **Policy / Scientific Question:** *How did consecutive MAM and OND seasons compound agricultural and livelihood shocks in the selected analogue year?*

6.5.1 Scientific Purpose & Context

Quantify and communicate Two-Season Compound Agricultural & Livelihood Outcome Matrix across Kenyan counties to inform anticipatory action and climate adaptation planning.

6.5.2 Mathematical Formulation & Algorithmic Approach

Baseline Reference Period: 1991–2020 WMO standard normal

Two-season compound agricultural impact table pairing OND and MAM analog seasons with observed crop and rangeland yields.

Applies non-parametric or standardized metrics aligned with Decision D43 and D53 to avoid distributional distortion.

6.5.3 Underlying Data Assets & Inputs

Provenance Key	Dataset Title	Spatial Resolution	Temporal Span
`enso-driver-indices`	Ocean Climate Teleconnection Drivers (Pa	global-oceans, equatorial-pacific, indian-ocean	1950-01 to present (open-ended)
`knbs-napr`	KNBS National Agricultural Production Re	kenya	2019 to 2024
`harveststat-crop-production`	HarvestStat Africa - Harmonized Subnatio	None	None

Served Parquet Files in Platform: `harveststat_county_production.parquet`, `driver_indices.parquet`

6.5.4 Methodological Assumptions & Analytical Limitations

- County aggregations rely on official polygon boundaries from UN OCHA COD-AB standard.
- Sub-seasonal temporal variations within rolling seasons are smoothed over seasonal totals.
- Data latency reflects standard publication cadences of underlying satellite and administrative surveys.

6.5.5 Controlled Vocabulary Keywords

`Compound Risk` • `Analog Seasons` • `Crop Yield`

6.6 Table 2.2: Official KMSA Seasonal Sectoral Advisory Matrix

Section in Platform: Section 2 | **Target Identifier:** `table_2_2` | **Output Type:** Table

> **Policy / Scientific Question:** *What anticipatory actions are recommended across agriculture, livestock, water, and disaster management sectors?*

6.6.1 Scientific Purpose & Context

Quantify and communicate Official KMSA Seasonal Sectoral Advisory Matrix across Kenyan counties to inform anticipatory action and climate adaptation planning.

6.6.2 Mathematical Formulation & Algorithmic Approach

Baseline Reference Period: 1991–2020 WMO standard normal

Tabular sectoral guidelines cross-referencing KMD forecast advisories across Agriculture, Water, Livestock, and Disaster Risk Reduction sectors.

Applies non-parametric or standardized metrics aligned with Decision D43 and D53 to avoid distributional distortion.

6.6.3 Underlying Data Assets & Inputs

Provenance Key	Dataset Title	Spatial Resolution	Temporal Span
`kmd-official-advisories`	kmd-official-advisories	County (ADM1)	Historical multi-decadal

Served Parquet Files in Platform: `kmd_cap_county_active.parquet`

6.6.4 Methodological Assumptions & Analytical Limitations

- County aggregations rely on official polygon boundaries from UN OCHA COD-AB standard.
- Sub-seasonal temporal variations within rolling seasons are smoothed over seasonal totals.
- Data latency reflects standard publication cadences of underlying satellite and administrative surveys.

6.6.5 Controlled Vocabulary Keywords

`KMD • Sectoral Guidelines • Anticipatory Action`

6.7 Table 2.3: Official Forecast, Early Warning & Advisory Directory for Kenya

Section in Platform: Section 2 | **Target Identifier:** `table_2_3` | **Output Type:** Table

> **Policy / Scientific Question:** *Where can county planners access official operational warnings and seasonal forecasts?*

6.7.1 Scientific Purpose & Context

Quantify and communicate Official Forecast, Early Warning & Advisory Directory for Kenya across Kenyan counties to inform anticipatory action and climate adaptation planning.

6.7.2 Mathematical Formulation & Algorithmic Approach

Baseline Reference Period: 1991–2020 WMO standard normal

Official early warning skill metrics evaluating Ranked Probability Skill Score (RPSS) and Leave-One-Out Cross-Validation (LOOCV) modal hit rates across lead times.

Applies non-parametric or standardized metrics aligned with Decision D43 and D53 to avoid distributional distortion.

Governing Mathematical Formulations:

$$\text{RPSS} = 1 - \frac{\text{RPS}_{\text{forecast}}}{\text{RPS}_{\text{climatology}}}$$

$$\text{Hit Rate} = \frac{1}{N} \sum_{i=1}^{N} \mathbb{I}(\hat{T}_i^{\text{modal}} == T_i^{\text{observed}})$$

6.7.3 Underlying Data Assets & Inputs

Provenance Key	Dataset Title	Spatial Resolution	Temporal Span
`enso-driver-indices`	Ocean Climate Teleconnection Drivers (Pa	global-oceans, equatorial-pacific, indian-ocean	1950-01 to present (open-ended)
`enso-hindcast-skill`	enso-hindcast-skill	County (ADM1)	Historical multi-decadal

Served Parquet Files in Platform: `enso_hindcast_skill.parquet`, `enso_state_probabilities.parq`

6.7.4 Methodological Assumptions & Analytical Limitations

- County aggregations rely on official polygon boundaries from UN OCHA COD-AB standard.
- Sub-seasonal temporal variations within rolling seasons are smoothed over seasonal totals.
- Data latency reflects standard publication cadences of underlying satellite and administrative surveys.

6.7.5 Controlled Vocabulary Keywords

`RPSS` • `LOOCV` • `Modal Hit Rate` • `Forecast Skill`

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7 Precipitation Climatology, Empirical Quantiles & Regimes

8 Precipitation Climatology, Empirical Quantiles & Regimes

This chapter defines the non-parametric empirical quantile baselines ($P_{33}, P_{67}, P_{10}, P_{90}$) over the 1991–2020 WMO standard normal period. It provides mathematical rationale for treating extreme positive rainfall skewness without Gaussian distortion, and formalizes independent seasonal standardization for consecutive sequence classification (Decision D43 & D53).

8.1 Figure 3.1: Multi-Decadal Precipitation Climatology & Standardized Anomalies

Section in Platform: Section 3 | **Target Identifier:** `fig_2_1` | **Output Type:** Chart

> **Policy / Scientific Question:** *How unusual was seasonal rainfall or drought across the historical satellite record?*

8.1.1 Scientific Purpose & Context

Quantify and communicate Multi-Decadal Precipitation Climatology & Standardized Anomalies across Kenyan counties to inform anticipatory action and climate adaptation planning.

8.1.2 Mathematical Formulation & Algorithmic Approach

Baseline Reference Period: 1991–2020 WMO standard normal

Non-parametric empirical quantiles (P33, P67, P10, P90) computed over the 1991–2020 WMO climatological baseline (Decision D43). Preserves positive rainfall skewness without Gaussian distortion.

Applies non-parametric or standardized metrics aligned with Decision D43 and D53 to avoid distributional distortion.

Governing Mathematical Formulations:

$$T_{33} = \hat{Q}_{1991\text{–}2020}(0.333), \quad T_{67} = \hat{Q}_{1991\text{–}2020}(0.667)$$

$$D_{10} = \hat{Q}_{1991\text{–}2020}(0.100), \quad D_{90} = \hat{Q}_{1991\text{–}2020}(0.900)$$

8.1.3 Underlying Data Assets & Inputs

Provenance Key	Dataset Title	Spatial Resolution	Temporal Span
`africa-precipitation-monthly-seasonal`	Monthly and Seasonal Precipitation from	africa — 0.05 deg (~5.5 km, CHIRPS native)	1981-01 to present (open-ended)

Served Parquet Files in Platform: `chirps_county.parquet`, `chirps_county_monthly.parquet`

8.1.4 Methodological Assumptions & Analytical Limitations

- County aggregations rely on official polygon boundaries from UN OCHA COD-AB standard.
- Sub-seasonal temporal variations within rolling seasons are smoothed over seasonal totals.
- Data latency reflects standard publication cadences of underlying satellite and administrative surveys.

8.1.5 Controlled Vocabulary Keywords

`CHIRPS v3` • `WMO Normal Baseline` • `Empirical Quantiles` • `Positive Skewness` • `Terciles`

8.2 Figure 3.1B: Consecutive Seasonal Rainfall Sequences & Compounding Drought Runs

Section in Platform: Section 3 | **Target Identifier:** `fig_2_1b` | **Output Type:** Chart

> **Policy / Scientific Question:** *Did this county's two rainy seasons fail back to back, how often has that happened before, and what happened when it did?*

8.2.1 Scientific Purpose & Context

Quantify and communicate Consecutive Seasonal Rainfall Sequences & Compounding Drought Runs across Kenyan counties to inform anticipatory action and climate adaptation planning.

8.2.2 Mathematical Formulation & Algorithmic Approach

Baseline Reference Period: 1991–2020 WMO standard normal

Consecutive seasonal sequence classification (Dry/Dry, Dry/Wet, Wet/Wet, Wet/Dry). Standardizes OND and MAM independently against each season's historical mean and standard deviation (Decision D53).

Applies non-parametric or standardized metrics aligned with Decision D43 and D53 to avoid distributional distortion.

Governing Mathematical Formulations:

$$z_{\text{OND}} = \frac{R_{\text{OND}} - \mu_{\text{OND}}}{\sigma_{\text{OND}}}, \quad z_{\text{MAM}} = \frac{R_{\text{MAM}} - \mu_{\text{MAM}}}{\sigma_{\text{MAM}}}$$

$$\text{Class}(s) = \begin{cases} \text{Dry} & \text{if } z \leq -0.5 \\ \text{Wet} & \text{if } z \geq +0.5 \\ \text{Normal} & \text{otherwise} \end{cases}$$

8.2.3 Underlying Data Assets & Inputs

Provenance Key	Dataset Title	Spatial Resolution	Temporal Span
`africa-precipitation-monthly-seasonal`	Monthly and Seasonal Precipitation from	africa — 0.05 deg (~5.5 km, CHIRPS native)	1981-01 to present (open-ended)

Served Parquet Files in Platform: `chirps_county_monthly.parquet`

8.2.4 Methodological Assumptions & Analytical Limitations

- County aggregations rely on official polygon boundaries from UN OCHA COD-AB standard.
- Sub-seasonal temporal variations within rolling seasons are smoothed over seasonal totals.
- Data latency reflects standard publication cadences of underlying satellite and administrative surveys.

8.2.5 Controlled Vocabulary Keywords

`Independent Standardization` • `Consecutive Sequence` • `Double Failure` • `Dry Run`

8.3 Figure 3.2: Empirical Contingency Probability Matrix

Section in Platform: Section 3 | **Target Identifier:** `fig_2_2` | **Output Type:** Chart

> **Policy / Scientific Question:** *What is the empirical probability of a wet, near-normal, or dry season given ocean driver phase?*

8.3.1 Scientific Purpose & Context

Quantify and communicate Empirical Contingency Probability Matrix across Kenyan counties to inform anticipatory action and climate adaptation planning.

8.3.2 Mathematical Formulation & Algorithmic Approach

Baseline Reference Period: 1991–2020 WMO standard normal

Joint empirical contingency matrix linking Pacific (RONI) and Indian Ocean (DMI) phases to observed rainfall tercile frequencies.

Applies non-parametric or standardized metrics aligned with Decision D43 and D53 to avoid distributional distortion.

Governing Mathematical Formulations:

$$P(T_k \mid \text{Phase}_j) = \frac{N(T_k \cap \text{Phase}_j)}{N(\text{Phase}_j)}$$

8.3.3 Underlying Data Assets & Inputs

Provenance Key	Dataset Title	Spatial Resolution	Temporal Span
`africa-precipitation-monthly-seasonal`	Monthly and Seasonal Precipitation from	africa — 0.05 deg (~5.5 km, CHIRPS native)	1981-01 to present (open-ended)
`enso-driver-indices`	Ocean Climate Teleconnection Drivers (Pa	global-oceans, equatorial-pacific, indian-ocean	1950-01 to present (open-ended)

Served Parquet Files in Platform: `chirps_county_monthly.parquet`, `driver_indices.parquet`

8.3.4 Methodological Assumptions & Analytical Limitations

- County aggregations rely on official polygon boundaries from UN OCHA COD-AB standard.
- Sub-seasonal temporal variations within rolling seasons are smoothed over seasonal totals.
- Data latency reflects standard publication cadences of underlying satellite and administrative surveys.

8.3.5 Controlled Vocabulary Keywords

`Contingency Matrix` • `Joint Ocean Phase` • `Terciles` • `RONI` • `DMI`

8.4 Figure 3.3: Oceanic Driver Teleconnection Relationships (OLS Regression)

Section in Platform: Section 3 | **Target Identifier:** `fig_2_3` | **Output Type:** Chart

> **Policy / Scientific Question:** *What is the continuous sensitivity of county precipitation to Pacific and Indian Ocean warming?*

8.4.1 Scientific Purpose & Context

Quantify and communicate Oceanic Driver Teleconnection Relationships (OLS Regression) across Kenyan counties to inform anticipatory action and climate adaptation planning.

8.4.2 Mathematical Formulation & Algorithmic Approach

Baseline Reference Period: 1991–2020 WMO standard normal

Continuous scatter correlation between standardized oceanic driver indices and seasonal rainfall anomalies with ordinary least squares (OLS) regression line and confidence bands.

Applies non-parametric or standardized metrics aligned with Decision D43 and D53 to avoid distributional distortion.

Governing Mathematical Formulations:

$$r = \frac{\sum(z_{\text{driver}} - \bar{z})(R - \bar{R})}{\sqrt{\sum(z_{\text{driver}} - \bar{z})^2 \sum(R - \bar{R})^2}}$$

8.4.3 Underlying Data Assets & Inputs

Provenance Key	Dataset Title	Spatial Resolution	Temporal Span
`africa-precipitation-monthly-seasonal`	Monthly and Seasonal Precipitation from	africa — 0.05 deg (~5.5 km, CHIRPS native)	1981-01 to present (open-ended)
`enso-driver-indices`	Ocean Climate Teleconnection Drivers (Pa	global-oceans, equatorial-pacific, indian-ocean	1950-01 to present (open-ended)

Served Parquet Files in Platform: `chirps_county_monthly.parquet`, `driver_indices.parquet`

8.4.4 Methodological Assumptions & Analytical Limitations

- County aggregations rely on official polygon boundaries from UN OCHA COD-AB standard.
- Sub-seasonal temporal variations within rolling seasons are smoothed over seasonal totals.
- Data latency reflects standard publication cadences of underlying satellite and administrative surveys.

8.4.5 Controlled Vocabulary Keywords

`Pearson Correlation` • `OLS Regression` • `Teleconnection Gradient`

8.5 Figure 3.4: Multi-Decadal Gridded Field History & Spatial Patterns

Section in Platform: Section 3 | **Target Identifier:** `fig_2_4` | **Output Type:** Map

> **Policy / Scientific Question:** *Where within the county did historical rainfall, vegetation, or drought anomalies concentrate?*

8.5.1 Scientific Purpose & Context

Quantify and communicate Multi-Decadal Gridded Field History & Spatial Patterns across Kenyan counties to inform anticipatory action and climate adaptation planning.

8.5.2 Mathematical Formulation & Algorithmic Approach

Baseline Reference Period: 1991–2020 WMO standard normal

Multi-decadal gridded rainfall raster analysis evaluating spatial gradients, elevation effects, and rainshadow patterns across subcounty zones.

Applies non-parametric or standardized metrics aligned with Decision D43 and D53 to avoid distributional distortion.

8.5.3 Underlying Data Assets & Inputs

Provenance Key	Dataset Title	Spatial Resolution	Temporal Span
`africa-precipitation-monthly-seasonal`	Monthly and Seasonal Precipitation from	africa — 0.05 deg (~5.5 km, CHIRPS native)	1981-01 to present (open-ended)

Served Parquet Files in Platform: `chirps_county.parquet`

8.5.4 Methodological Assumptions & Analytical Limitations

- County aggregations rely on official polygon boundaries from UN OCHA COD-AB standard.
- Sub-seasonal temporal variations within rolling seasons are smoothed over seasonal totals.
- Data latency reflects standard publication cadences of underlying satellite and administrative surveys.

8.5.5 Controlled Vocabulary Keywords

`CHIRPS Gridded • Spatial Gradient • Elevation Influence`

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9 Satellite Earth Observation, Hazards & Exposure

10 Satellite Earth Observation, Hazards & Exposure

This chapter covers Earth observation satellite telemetry for rapid flood inundation and rangeland vegetative health. It formalizes Sentinel-1 Synthetic Aperture Radar (SAR) backscatter thresholding from the Copernicus Global Flood Monitoring (GFM) service, hydrodynamic return period modeling (JRC), and MODIS 250m dekadal NDVI vegetation greenness anomalies.

10.1 Figure 3.5: Subcounty Flood Hazard & Infrastructure Exposure Breakdown

Section in Platform: Section 3 | **Target Identifier:** `fig_2_5` | **Output Type:** Map

> **Policy / Scientific Question:** *Which subcounties, populations, and public infrastructure assets face extreme flood inundation?*

10.1.1 Scientific Purpose & Context

Quantify and communicate Subcounty Flood Hazard & Infrastructure Exposure Breakdown across Kenyan counties to inform anticipatory action and climate adaptation planning.

10.1.2 Mathematical Formulation & Algorithmic Approach

Baseline Reference Period: 1991–2020 WMO standard normal

Subcounty flood hazard profiling pairing Copernicus GFM Sentinel-1 SAR observed surface water extent with JRC flood return periods (1-in-10 to 1-in-100 year models).

Applies non-parametric or standardized metrics aligned with Decision D43 and D53 to avoid distributional distortion.

Governing Mathematical Formulations:

$$\text{Exposure Rate} = \frac{\text{Pop}_{\text{inundated}}}{\text{Pop}_{\text{total}}} \times 100\%$$

10.1.3 Underlying Data Assets & Inputs

Provenance Key	Dataset Title	Spatial Resolution	Temporal Span
`kenya-flood-gfm`	Kenya Observed Flood Extent (Copernicus	kenya — 0.001 deg (~111 m); aggregated from 20 m Sentinel-1 tiles (a coarse pixel is flooded if any 20 m sub-pixel flooded)	2018-01 to present (open-ended)
`eastafrica-flood-jrc`	JRC / GloFAS Return-Period River Flood H	world, kenya — 3 arc-seconds (~90 m)	None
`kenya-facilities-hotosm`	Kenya Health and Education Facilities (H	kenya — Facility features (points, some building polygons)	2026-08
`kenya-roads-osm`	Kenya Classified Road Network (OpenStree	kenya	2026-08

Served Parquet Files in Platform: `exposure_gfm_seasonal.parquet`, `exposure_jrc_rp.parquet`

10.1.4 Methodological Assumptions & Analytical Limitations

- County aggregations rely on official polygon boundaries from UN OCHA COD-AB standard.
- Sub-seasonal temporal variations within rolling seasons are smoothed over seasonal totals.
- Data latency reflects standard publication cadences of underlying satellite and administrative surveys.

10.1.5 Controlled Vocabulary Keywords

`Copernicus GFM` • `Sentinel-1 SAR` • `JRC Flood RP` • `Return Period`

10.2 Figure 4.3: Continuous MODIS Dekadal NDVI Pasture Biomass Dynamics

Section in Platform: Section 4 | **Target Identifier:** `fig_3_2` | **Output Type:** Chart

Policy / Scientific Question: *When did historical rangeland vegetation collapse into severe pasture deficits?*

10.2.1 Scientific Purpose & Context

Quantify and communicate Continuous MODIS Dekadal NDVI Pasture Biomass Dynamics across Kenyan counties to inform anticipatory action and climate adaptation planning.

10.2.2 Mathematical Formulation & Algorithmic Approach

Baseline Reference Period: 1991–2020 WMO standard normal

Continuous MODIS 250m 16-day NDVI vegetation anomaly z-scores tracking pasture biomass, forage desiccation, and post-shock rangeland green-up.

Applies non-parametric or standardized metrics aligned with Decision D43 and D53 to avoid distributional distortion.

Governing Mathematical Formulations:

$$z_{\mathrm{NDVI},t} = \frac{\mathrm{NDVI}_t - \mathrm{Median}_t(2000\text{–}2020)}{\sigma_t(2000\text{–}2020)}$$

10.2.3 Underlying Data Assets & Inputs

Provenance Key	Dataset Title	Spatial Resolution	Temporal Span
`eastafrica-ndvi-modis`	MODIS NDVI Seasonal Vegetation Condition	eastern-africa, kenya — ~250 m (0.0021 deg, MOD13Q1 native)	2000 to 2025
`enso-driver-indices`	Ocean Climate Teleconnection Drivers (Pa	global-oceans, equatorial-pacific, indian-ocean	1950-01 to present (open-ended)

Served Parquet Files in Platform: `ndvi_county.parquet`, `driver_indices.parquet`

10.2.4 Methodological Assumptions & Analytical Limitations

- County aggregations rely on official polygon boundaries from UN OCHA COD-AB standard.
- Sub-seasonal temporal variations within rolling seasons are smoothed over seasonal totals.
- Data latency reflects standard publication cadences of underlying satellite and administrative surveys.

10.2.5 Controlled Vocabulary Keywords

`MODIS NDVI` • `Pasture Biomass` • `Greenness Anomaly` • `Rangeland Recovery`

10.3 Figure 4.5: Subcounty Sentinel-1 SAR Flood Inundation & Asset Exposure

Section in Platform: Section 4 | **Target Identifier:** `fig_3_3` | **Output Type:** Map

> **Policy / Scientific Question:** *What was the observed satellite footprint of extreme flood events and what assets were exposed?*

10.3.1 Scientific Purpose & Context

Quantify and communicate Subcounty Sentinel-1 SAR Flood Inundation & Asset Exposure across Kenyan counties to inform anticipatory action and climate adaptation planning.

10.3.2 Mathematical Formulation & Algorithmic Approach

Baseline Reference Period: 1991–2020 WMO standard normal

Spatial intersection model evaluating physical asset exposure to observed flood inundation: population headcount, cropland area, road network kilometers, and health/education facilities.

Applies non-parametric or standardized metrics aligned with Decision D43 and D53 to avoid distributional distortion.

Governing Mathematical Formulations:

$$E_{\text{asset}} = \int_{\Omega_{\text{inundated}}} D_{\text{asset}}(x, y)\, dxdy$$

Provenance Key	Dataset Title	Spatial Resolution	Temporal Span

10.3.3 Underlying Data Assets & Inputs

Provenance Key	Dataset Title	Spatial Resolution	Temporal Span
`kenya-flood-exposure-intersect`	Kenya Flood x Exposure, Per-Sub-County T	kenya — IEBC sub-counties (290 admin-2 units)	2018 to 2025
`kenya-flood-gfm`	Kenya Observed Flood Extent (Copernicus	kenya — 0.001 deg (~111 m); aggregated from 20 m Sentinel-1 tiles (a coarse pixel is flooded if any 20 m sub-pixel flooded)	2018-01 to present (open-ended)
`kenya-population-worldpop`	WorldPop Constrained Gridded Population	kenya — ~100 m (3 arc-seconds)	2020
`kenya-facilities-hotosm`	Kenya Health and Education Facilities (H	kenya — Facility features (points, some building polygons)	2026-08
`kenya-roads-osm`	Kenya Classified Road Network (OpenStree	kenya	2026-08

Served Parquet Files in Platform: `exposure_totals.parquet`, `exposure_vop.parquet`

10.3.4 Methodological Assumptions & Analytical Limitations

- County aggregations rely on official polygon boundaries from UN OCHA COD-AB standard.
- Sub-seasonal temporal variations within rolling seasons are smoothed over seasonal totals.
- Data latency reflects standard publication cadences of underlying satellite and administrative surveys.

10.3.5 Controlled Vocabulary Keywords

`Spatial Overlay` • `Inundation Exposure` • `Critical Infrastructure`

10.4 Figure 4.6: Humanitarian Emergency Chronology & Situation Reports

Section in Platform: Section 4 | **Target Identifier:** `fig_3_4` | **Output Type:** Chart

> **Policy / Scientific Question:** *What humanitarian crisis reports were filed by UN OCHA and government agencies across historical disasters?*

10.4.1 Scientific Purpose & Context

Quantify and communicate Humanitarian Emergency Chronology & Situation Reports across Kenyan counties to inform anticipatory action and climate adaptation planning.

10.4.2 Mathematical Formulation & Algorithmic Approach

Baseline Reference Period: 1991–2020 WMO standard normal

Humanitarian crisis impact chronology compiling UN OCHA ReliefWeb disaster dispatches and NDMA monthly drought early warning bulletins across alarm phases.

Applies non-parametric or standardized metrics aligned with Decision D43 and D53 to avoid distributional distortion.

10.4.3 Underlying Data Assets & Inputs

Provenance Key	Dataset Title	Spatial Resolution	Temporal Span
`reliefweb`	reliefweb	County (ADM1)	Historical multi-decadal
`ndma-bulletin-index`	ndma-bulletin-index	County (ADM1)	Historical multi-decadal

Served Parquet Files in Platform: `reliefweb_county.parquet`, `ndma_bulletin_index.parquet`

10.4.4 Methodological Assumptions & Analytical Limitations

- County aggregations rely on official polygon boundaries from UN OCHA COD-AB standard.
- Sub-seasonal temporal variations within rolling seasons are smoothed over seasonal totals.
- Data latency reflects standard publication cadences of underlying satellite and administrative surveys.

10.4.5 Controlled Vocabulary Keywords

ReliefWeb • NDMA Bulletin • Disaster Chronology • Drought Phase

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11 Agricultural Production & Economic Valuation

12 Agricultural Production & Economic Valuation

This chapter formalizes the agricultural production anomaly functions, historical crop yield detrending, and the county economic capital valuation engine. It documents KNBS National Agriculture Production Report (NAPR) extractions, HarvestStat historical yields, and the subcounty spatial weight apportionment model under Decision D62/D63 Beta review governance.

12.1 Figure 1.1: County Agricultural & Livestock Production Time Series (2019–2024)

Section in Platform: Section 1.1 | **Target Identifier:** `fig_1_1` | **Output Type:** Chart

> **Policy / Scientific Question:** *How has crop and livestock production varied over recent statutory reporting years?*

12.1.1 Scientific Purpose & Context

Quantify and communicate County Agricultural & Livestock Production Time Series (2019–2024) across Kenyan counties to inform anticipatory action and climate adaptation planning.

12.1.2 Mathematical Formulation & Algorithmic Approach

Baseline Reference Period: 1991–2020 WMO standard normal

Official agricultural and pastoral asset inventories from KNBS National Agriculture Production Report (2024/2025). County share computed relative to national 47-county totals.

Applies non-parametric or standardized metrics aligned with Decision D43 and D53 to avoid distributional distortion.

Governing Mathematical Formulations:

$$\text{Share}_c = \frac{Y_c}{\sum_{i=1}^{47} Y_i} \times 100\%$$

12.1.3 Underlying Data Assets & Inputs

Provenance Key	Dataset Title	Spatial Resolution	Temporal Span
`knbs-napr`	KNBS National Agricultural Production Re	kenya	2019 to 2024

Served Parquet Files in Platform: `knbs_napr_county_production.parquet`, `knbs_napr_livestock.parquet`, `knbs_napr_livestock_products.parquet`

12.1.4 Methodological Assumptions & Analytical Limitations

- County aggregations rely on official polygon boundaries from UN OCHA COD-AB standard.
- Sub-seasonal temporal variations within rolling seasons are smoothed over seasonal totals.
- Data latency reflects standard publication cadences of underlying satellite and administrative surveys.

12.1.5 Controlled Vocabulary Keywords

`KNBS NAPR` • `Crops` • `Livestock Headcount` • `Milk Production`

12.2 Figure 1.2: Modeled Value of Production Portfolio (MapSPAM & GLW4)

Section in Platform: Section 1.2 | **Target Identifier:** `fig_1_2` | **Output Type:** Chart

> **Policy / Scientific Question:** *What agricultural and livestock capital assets are at economic risk across the county?*

12.2.1 Scientific Purpose & Context

Quantify and communicate Modeled Value of Production Portfolio (MapSPAM & GLW4) across Kenyan counties to inform anticipatory action and climate adaptation planning.

12.2.2 Mathematical Formulation & Algorithmic Approach

Baseline Reference Period: 1991–2020 WMO standard normal

Multi-tier economic capital valuation modeling crop Gross Value of Production (VOP) in constant Int$2015 / USD and livestock herd capital replacement value. Subcounty apportionment via landcover and rangeland weighting (Decision D62/D63 Beta review governance).

Applies non-parametric or standardized metrics aligned with Decision D43 and D53 to avoid distributional distortion.

Governing Mathematical Formulations:

$$V_c = \sum_k Q_{c,k} \cdot P_k^{\text{Int\$}}$$

$$V_{sc} = V_c \cdot w_{sc}, \quad \sum_{sc \in c} w_{sc} = 1.0$$

12.2.3 Underlying Data Assets & Inputs

Provenance Key	Dataset Title	Spatial Resolution	Temporal Span
`crop-vop-intld15`	MapSPAM 2020 Adaptation Atlas, Sub-Sahar	sub-saharan-africa	2020
`livestock-vop`	Gridded Livestock of the World (GLW4) Va	sub-saharan-africa, kenya — 1 km (0.008333 deg)	2020
`knbs-napr`	KNBS National Agricultural Production Re	kenya	2019 to 2024

Served Parquet Files in Platform: `economic_valuation_baseline.parquet`, `economic_valuation_subcounty_weights.parquet`

12.2.4 Methodological Assumptions & Analytical Limitations

- County aggregations rely on official polygon boundaries from UN OCHA COD-AB standard.
- Sub-seasonal temporal variations within rolling seasons are smoothed over seasonal totals.
- Data latency reflects standard publication cadences of underlying satellite and administrative surveys.

12.2.5 Controlled Vocabulary Keywords

`VOP` • `Herd Capital` • `Subcounty Weights` • `Beta Review Governance`

12.3 Figure 4.1: Dual-Season Teleconnection Drivers vs KNBS Annual Output

Section in Platform: Section 4 | **Target Identifier:** `fig_3_1` | **Output Type:** Chart

> **Policy / Scientific Question:** *How do consecutive MAM and OND climate shocks translate into annual crop and livestock output shocks?*

12.3.1 Scientific Purpose & Context

Quantify and communicate Dual-Season Teleconnection Drivers vs KNBS Annual Output across Kenyan counties to inform anticipatory action and climate adaptation planning.

12.3.2 Mathematical Formulation & Algorithmic Approach

Baseline Reference Period: 1991–2020 WMO standard normal

Dual-season teleconnection response of KNBS crop yields and livestock parameters across El Niño, La Niña, and Neutral years. Detrended relative to baseline yields.

Applies non-parametric or standardized metrics aligned with Decision D43 and D53 to avoid distributional distortion.

Governing Mathematical Formulations:

$$\Delta Y_{\%} = \frac{Y - Y_{\text{trend}}}{Y_{\text{trend}}} \times 100\%$$

12.3.3 Underlying Data Assets & Inputs

Provenance Key	Dataset Title	Spatial Resolution	Temporal Span
`knbs-napr`	KNBS National Agricultural Production Re	kenya	2019 to 2024
`enso-driver-indices`	Ocean Climate Teleconnection Drivers (Pa	global-oceans, equatorial-pacific, indian-ocean	1950-01 to present (open-ended)

Served Parquet Files in Platform: `knbs_napr_county_production.parquet`, `knbs_napr_livestock.parquet`, `driver_indices.parquet`

12.3.4 Methodological Assumptions & Analytical Limitations

- County aggregations rely on official polygon boundaries from UN OCHA COD-AB standard.
- Sub-seasonal temporal variations within rolling seasons are smoothed over seasonal totals.
- Data latency reflects standard publication cadences of underlying satellite and administrative surveys.

12.3.5 Controlled Vocabulary Keywords

`Yield Anomaly` • `Detrended Yield` • `ENSO Phase Response`

12.4 Figure 4.2: Empirical Ocean Driver Phase Response vs Mean Output

Section in Platform: Section 4 | **Target Identifier:** `fig_3_1b` | **Output Type:** Chart

> **Policy / Scientific Question:** *What is the historical average production anomaly during El Niño, La Niña, or IOD phases?*

12.4.1 Scientific Purpose & Context

Quantify and communicate Empirical Ocean Driver Phase Response vs Mean Output across Kenyan counties to inform anticipatory action and climate adaptation planning.

12.4.2 Mathematical Formulation & Algorithmic Approach

Baseline Reference Period: 1991–2020 WMO standard normal

Empirical phase response distribution of agricultural production showing distribution spreads, interquartile ranges, and outlier seasons.

Applies non-parametric or standardized metrics aligned with Decision D43 and D53 to avoid distributional distortion.

12.4.3 Underlying Data Assets & Inputs

Provenance Key	Dataset Title	Spatial Resolution	Temporal Span
`knbs-napr`	KNBS National Agricultural Production Re	kenya	2019 to 2024
`enso-driver-indices`	Ocean Climate Teleconnection Drivers (Pa	global-oceans, equatorial-pacific, indian-ocean	1950-01 to present (open-ended)

Served Parquet Files in Platform: `knbs_napr_county_production.parquet`, `driver_indices.parquet`

12.4.4 Methodological Assumptions & Analytical Limitations

- County aggregations rely on official polygon boundaries from UN OCHA COD-AB standard.
- Sub-seasonal temporal variations within rolling seasons are smoothed over seasonal totals.
- Data latency reflects standard publication cadences of underlying satellite and administrative surveys.

12.4.5 Controlled Vocabulary Keywords

`Interquartile Range` • `Production Spread` • `Surplus` • `Deficit`

12.5 Figure 4.2B: Multi-Decadal Seasonal Crop Harvests & Climate Teleconnections (HarvestStat Africa)

Section in Platform: Section 4 | **Target Identifier:** `fig_3_1c` | **Output Type:** Chart

> **Policy / Scientific Question:** *How did subnational seasonal crop yields and production track historical rainfall terciles across 1990–2024?*

12.5.1 Scientific Purpose & Context

Quantify and communicate Multi-Decadal Seasonal Crop Harvests & Climate Teleconnections (HarvestStat Africa) across Kenyan counties to inform anticipatory action and climate adaptation planning.

12.5.2 Mathematical Formulation & Algorithmic Approach

Baseline Reference Period: 1991–2020 WMO standard normal

Long-term historical crop yield time-series (1982–2020 HarvestStat) detrended using linear and polynomial models to evaluate teleconnection persistence over four decades.

Applies non-parametric or standardized metrics aligned with Decision D43 and D53 to avoid distributional distortion.

Governing Mathematical Formulations:

$$Y_{\text{detrended}} = Y_t - (a \cdot t + b) + \bar{Y}$$

12.5.3 Underlying Data Assets & Inputs

Provenance Key	Dataset Title	Spatial Resolution	Temporal Span
`harveststat-crop-production`	HarvestStat Africa - Harmonized Subnatio	None	None
`enso-driver-indices`	Ocean Climate Teleconnection Drivers (Pa	global-oceans, equatorial-pacific, indian-ocean	1950-01 to present (open-ended)

Served Parquet Files in Platform: `harveststat_county_production.parquet`, `driver_indices.parquet`

12.5.4 Methodological Assumptions & Analytical Limitations

- County aggregations rely on official polygon boundaries from UN OCHA COD-AB standard.
- Sub-seasonal temporal variations within rolling seasons are smoothed over seasonal totals.
- Data latency reflects standard publication cadences of underlying satellite and administrative surveys.

12.5.5 Controlled Vocabulary Keywords

`HarvestStat` • `Decadal Persistence` • `Detrending`

12.6 Figure 4.4: Pastoral Market Prices & Terms of Trade Monthly Purchasing Power

Section in Platform: Section 4 | **Target Identifier:** `fig_3_2b` | **Output Type:** Chart

> **Policy / Scientific Question:** *How severely did food prices surge and pastoral purchasing power collapse during climate shocks?*

12.6.1 Scientific Purpose & Context

Quantify and communicate Pastoral Market Prices & Terms of Trade Monthly Purchasing Power across Kenyan counties to inform anticipatory action and climate adaptation planning.

12.6.2 Mathematical Formulation & Algorithmic Approach

Baseline Reference Period: 1991–2020 WMO standard normal

Pastoral terms-of-trade (ToT) computed as the ratio of live medium-sized goat (shoat) price to 1 kg dry maize grain. Spikes and collapses indexed against ENSO shock windows.

Applies non-parametric or standardized metrics aligned with Decision D43 and D53 to avoid distributional distortion.

Governing Mathematical Formulations:

$$\mathrm{ToT}_m = \frac{P_{\mathrm{goat},m}}{P_{\mathrm{maize},m}} \quad \text{(kg maize per goat)}$$

12.6.3 Underlying Data Assets & Inputs

Provenance Key	Dataset Title	Spatial Resolution	Temporal Span
`kenya-market-prices-fews`	Kenya Retail and Wholesale Market Prices	kenya	2000 to present (open-ended)
`kenya-market-prices-kamis`	Kenya Live-Animal Market Prices by Grade	kenya	2021-05-24 to present (open-ended)

Served Parquet Files in Platform: `market_prices.parquet`, `market_prices_kamis.parquet`

12.6.4 Methodological Assumptions & Analytical Limitations

- County aggregations rely on official polygon boundaries from UN OCHA COD-AB standard.
- Sub-seasonal temporal variations within rolling seasons are smoothed over seasonal totals.
- Data latency reflects standard publication cadences of underlying satellite and administrative surveys.

12.6.5 Controlled Vocabulary Keywords

`Terms of Trade` • `Shoat Price` • `Maize Grain` • `Purchasing Power`

12.7 Figure 4.4B: Regional Cross-Border Food Trade Flows & Market Shock Absorber (FEWS NET XBT)

Section in Platform: Section 4 | **Target Identifier:** `fig_3_2c` | **Output Type:** Chart

> **Policy / Scientific Question:** *How do regional food trade flows from Uganda, Tanzania, and Ethiopia buffer domestic grain supply during rainfall shocks?*

12.7.1 Scientific Purpose & Context

Quantify and communicate Regional Cross-Border Food Trade Flows & Market Shock Absorber (FEWS NET XBT) across Kenyan counties to inform anticipatory action and climate adaptation planning.

12.7.2 Mathematical Formulation & Algorithmic Approach

Baseline Reference Period: 1991–2020 WMO standard normal

East Africa cross-border formal and informal grain trade flows (FEWS NET XBT). Evaluates supply buffering from Uganda and Tanzania during regional drought shocks.

Applies non-parametric or standardized metrics aligned with Decision D43 and D53 to avoid distributional distortion.

12.7.3 Underlying Data Assets & Inputs

Provenance Key	Dataset Title	Spatial Resolution	Temporal Span
`xbt-trade`	xbt-trade	County (ADM1)	Historical multi-decadal

Served Parquet Files in Platform: `xbt_trade.parquet`

12.7.4 Methodological Assumptions & Analytical Limitations

- County aggregations rely on official polygon boundaries from UN OCHA COD-AB standard.
- Sub-seasonal temporal variations within rolling seasons are smoothed over seasonal totals.
- Data latency reflects standard publication cadences of underlying satellite and administrative surveys.

12.7.5 Controlled Vocabulary Keywords

`XBT Trade` • `Cross-Border Trade` • `Supply Buffering`

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13 Master Dataset Catalogue & Data Provenance

14 Master Dataset Catalogue & Data Provenance

This chapter documents the 23 formal data assets serving the Kenya ENSO Explorer platform. Each entry conforms to the Comprehensive Data Hub (CDH) Schema v0.3.0, recording upstream producers, spatial-temporal coverages, scientific caveats, and open-access licensing terms.

14.1 `enso-driver-indices`: Ocean Climate Teleconnection Drivers (Pacific RONI, Indian Ocean DMI & Western-V)

Category: Climate | **Publishing State:** `authored` | **License:** Open Data

14.1.1 Summary

Harmonised multi-decadal monthly and seasonal time series of primary equatorial ocean-atmosphere climate teleconnection drivers governing Kenyan rainfall: NOAA CPC Relative Oceanic Niño Index (RONI) for Pacific ENSO state; NOAA CPC / HadISST Dipole Mode Index (DMI) for Indian Ocean Dipole state; and Funk et al. Western-V SST index for MAM Long Rains teleconnections. Standardised and compiled into analysis-ready parquet tables (1950 to present).

- **Temporal Coverage:** 1950-01 to present (open-ended)
- **Spatial Resolution:** global-oceans, equatorial-pacific, indian-ocean

14.1.2 Data Producers & Custodians

- **NOAA Climate Prediction Center** - https://www.cpc.ncep.noaa.gov/
- **Alliance of Bioversity International and CIAT** (Peter Steward)

14.1.3 Scientific Caveats & Guidance

[!WARNING] Attribution: “Data provided by NOAA Climate Prediction Center (CPC) and compiled for the African Agriculture Adaptation Atlas (AAAA).” RELATIVE OCEANIC NINO INDEX (RONI) OVER RAW NINO 3.4. CPC computes RONI by subtracting the tropical-mean (20S-20N) SST anomaly from the Nino 3.4 anomaly. This detrending removes secular tropical ocean warming (~+0.6 degC since 1950) that otherwise causes historical cool baselines to register as artificial permanent El Nino conditions. Always use RONI for multi-decadal analogue selection. INDIAN OCEAN DIPOLE (IOD) DOMINANCE IN SHORT RAINS. The DMI is the primary physical forcing of Kenyan rainfall variability during the October-December (OND) Short Rains. Warm ENSO and positive IOD co-occur frequently but can decouple; off-diagonal states (El Nino + negative IOD) produce subdued anomalies. SEASONAL ASYMMETRY AND LONG RAINS (MAM) DECOUPLING. Pacific ENSO indices exhibit weak correlation with Kenyan rainfall during the March-May (MAM) Long Rains (r ~ 0.08). The Western-V index captures western Pacific warm pool gradients that govern convective subsidence over the Horn of Africa, providing independent predictive skill where ENSO fails. Z-SCORE STANDARDIZATION REQUIREMENT. Pacific SST anomalies (~0.5-2.5 degC) and regional rainfall anomalies exhibit a 10.6:1 variance distortion ratio. Multi-driver analogue selection must use z-score standardisation across all driver dimensions to ensure equal physical weighting.

14.2 `africa-precipitation-monthly-seasonal`: Monthly and Seasonal Precipitation from CHIRPS v3 (Africa)

Category: Climate | **Publishing State:** `authored` | **License:** Open Data

14.2.1 Summary

Total precipitation (PTOT) aggregated from CHIRPS v3.0 and delivered as per-pixel monthly Cloud Optimized GeoTIFFs and per-year rolling 3-month (seasonal) sums over Africa, 1981 to present. This is the rainfall backbone of the KE-ENSO Explorer’s seasonal/ENSO analysis: the twelve seasonal windows (JFM, FMA, …, DJF) are the tri-month sums the notebook compares against ENSO phase and against the drought (SPEI), vegetation (NDVI) and flood layers. A derived aggregation product, not a redistribution of the CHIRPS daily archive.

- **Temporal Coverage:** 1981-01 to present (open-ended)
- **Spatial Resolution:** africa — 0.05 deg (~5.5 km, CHIRPS native)

14.2.2 Data Producers & Custodians

- **Alliance of Bioversity International and CIAT** (Peter Steward)
- **Climate Hazards Center, UC Santa Barbara** - https://www.chc.ucsb.edu/data/chirps

14.2.3 Scientific Caveats & Guidance

> [!WARNING] Source CHIRPS is public domain ("no restrictions on use"); these reprocessed COGs are released CC-BY-4.0 with attribution to the Climate Hazards Center, UC Santa Barbara. Seasonal windows are labelled by the calendar year in which the window ends (DJF_2020 = Dec 2019 to Feb 2020). Edge windows that the record cannot cover in full are absent (NDJ and DJF 1981; windows ending after the latest monthly file), so not every season x year combination of the template exists.

14.3 `africa-spei-drought`: SPEI Drought Index, 3- and 12-month (Africa)

Category: Climate | **Publishing State:** `authored` | **License:** Open Data

14.3.1 Summary

Standardized Precipitation-Evapotranspiration Index at 3-month (SPEI-03) and 12-month (SPEI-12) accumulation, as per-pixel monthly Cloud Optimized GeoTIFFs over Africa, 1981 to present. Built from CHIRPS v3 precipitation and a potential-evapotranspiration climate water balance fitted with a log-logistic distribution; negative values are drier than the reference climatology. The drought layer of the KE-ENSO Explorer: the short accumulation tracks seasonal/agricultural drought, the long accumulation hydrological drought.

- **Temporal Coverage:** 1981-01 to present (open-ended)
- **Spatial Resolution:** africa — 0.05 deg (~5.5 km, CHIRPS grid)

14.3.2 Data Producers & Custodians

- **Alliance of Bioversity International and CIAT** (Peter Steward)

14.3.3 Scientific Caveats & Guidance

[!WARNING] SPEI is a z-score anomaly against the fitted reference distribution, not an absolute deficit. Arid-cell -Inf values (zero-precipitation tails of the log-logistic fit) are clamped to NoData.

14.4 `eastafrica-flood-jrc`: JRC / GloFAS Return-Period River Flood Hazard (Global)

Category: Satellite | **Publishing State:** `authored` | **License:** Open Data

14.4.1 Summary

Modelled global riverine flood hazard from the JRC / GloFAS Global Flood Maps (CEMS-GLOFAS v2.1.2): inundation depth for seven return periods, 10 to 500 years, at 3 arc-seconds, as a static baseline climatology. Published as Cloud Optimized GeoTIFFs with internal overviews on Source Cooperative, which is the canonical read location. The Adaptation Atlas additionally holds a provisional Kenya crop of the same data. Within the KE-ENSO Explorer this is the MODELLED flood-prone baseline complementing the OBSERVED Copernicus GFM extent (record kenya-flood-gfm): JRC answers "how flood-prone at magnitude X", GFM answers "where and when it actually flooded".

- **Temporal Coverage:** None
- **Spatial Resolution:** world, kenya — 3 arc-seconds (~90 m)

14.4.2 Data Producers & Custodians

- **Alliance of Bioversity International and CIAT** (Peter Steward)
- **Source Cooperative / Radiant Earth Foundation** - https://source.coop/nlebovits/jrc-glofas
- **European Commission Joint Research Centre (JRC)** - https://data.jrc.ec.europa.eu/

14.4.3 Scientific Caveats & Guidance

[!WARNING] FEDERATED TO THE CLOUD-OPTIMIZED UPSTREAM (2026-09-18, following review on cdh-catalog#32). The full global product is already published as COGs with internal overviews under CC-BY-4.0 on Source Cooperative, at the same seven return periods and the same 3 arc-second resolution as the Atlas crop. The canonical asset in this record is therefore the global collection, not our subset. The Kenya crop is retained as a provisional second asset because

the KE-ENSO Explorer currently reads it, but it carries no information the global collection lacks and the performance argument for keeping it is untested. Measuring that is the open task; retiring the duplicate is the likely outcome. Static modelled hazard (return-period depths); not an observed time series and not scenario-projected, so no temporal coverage is declared. Pair with the GFM observed extent for the "modelled vs observed" flood story.

14.5 `kenya-flood-gfm`: Kenya Observed Flood Extent (Copernicus GFM, Sentinel-1)

Category: Satellite | **Publishing State:** `authored` | **License:** Open Data

14.5.1 Summary

Observed flood extent for Kenya from the Copernicus Emergency Management Service Global Flood Monitoring (GFM) product, which classifies flooding from every Sentinel-1 SAR acquisition (ensemble of the DLR, TU Wien and LIST algorithms; permanent and reference water excluded). Delivered as ~111 m aggregates built directly from the 20 m SAR tiles: a monthly flood-occurrence layer with its valid-observation count, a rolling 3-month (seasonal) layer aligned to the KE-ENSO Explorer precipitation seasons, and a full-record history (occurrence frequency and ever-flooded footprint), 2018 to present. Replaces the MODIS Global Flood Database (ended 2018) with a cloud-penetrating, continuously updated SAR record. The 20 m per-acquisition archive is not materialised here (16,000+ acquisitions); it can be re-pulled from the EODC STAC on demand.

- **Temporal Coverage:** 2018-01 to present (open-ended)
- **Spatial Resolution:** kenya — 0.001 deg (~111 m); aggregated from 20 m Sentinel-1 tiles (a coarse pixel is flooded if any 20 m sub-pixel flooded)

14.5.2 Data Producers & Custodians

- **Alliance of Bioversity International and CIAT** (Peter Steward)
- **Copernicus Emergency Management Service (CEMS)** - https://global-flood.emergency.copernicus.eu/

14.5.3 Scientific Caveats & Guidance

[!WARNING] Attribution required: “Contains modified Copernicus Emergency Management Service information [year]”. Pixel coding of the flooded layers: 0 = not flooded, 1 = flooded, 255 = not observed (SAR gap or excluded) - 255 is absence of observation, NOT dry; the nobs layer gives the count of valid observations so no-flood is distinguishable from no-observation. Seasonal windows are named by the initials of their months and labelled by the year of their FIRST month (DJF_2018 = Dec 2018 to Feb 2019), so NDJ and DJF exist for 2018 to 2024 only. This differs from the CHIRPS seasonal record (africa-precipitation-monthly-seasonal), which labels NDJ and DJF by the year the window ends; align on those two windows with care. A ~7-year record measures recent observed flood occurrence, not return-period magnitude.

14.6 `eastafrica-ndvi-modis`: MODIS NDVI Seasonal Vegetation Condition (East Africa)

Category: Satellite | **Publishing State:** `authored` | **License:** Open Data

14.6.1 Summary

Normalized Difference Vegetation Index from MODIS MOD13Q1 (250 m, 16-day composites), averaged over the two East African rainy seasons - the long rains (MAM, March to May) and the short rains (OND, October to December) - for every year from 2000 to 2025, over an East Africa window centred on Kenya. Written as Cloud Optimized GeoTIFFs. The vegetation-condition layer of the KE-ENSO Explorer: greenness response to seasonal rainfall and ENSO phase.

- **Temporal Coverage:** 2000 to 2025
- **Spatial Resolution:** eastern-africa, kenya — ~250 m (0.0021 deg, MOD13Q1 native)

14.6.2 Data Producers & Custodians

- **Alliance of Bioversity International and CIAT** (Peter Steward)
- **NASA LP DAAC / USGS EROS** - https://lpdaac.usgs.gov/products/mod13q1v061/

14.6.3 Scientific Caveats & Guidance

[!WARNING] Source MODIS data are free and unrestricted; these aggregated COGs are released CC-BY-4.0 with attribution to NASA LP DAAC. NDVI ranges roughly -0.2 to 1.0 (higher = greener). Only the MAM and OND windows are published (no other seasons, no annual mean). Each file covers the full East Africa window (~275 MB); read with a bbox/overview to avoid whole-file downloads.

14.7 `eastafrica-wrsi-fews`: FEWS NET Water Requirement Satisfaction Index, Cropland and Rangeland (Kenya)

Category: Satellite | **Publishing State:** `authored` | **License:** Open Data

14.7.1 Summary

End-of-season Water Requirement Satisfaction Index (WRSI) for cropland (maize) and rangeland growing seasons from USGS FEWS NET, for the long rains (MAM) and short rains (OND) of every year from 2003, cropped to Kenya and written as Cloud Optimized GeoTIFFs. WRSI is the ratio of seasonal water supplied to crop or pasture water demand (0 to 100 percent); low values flag agricultural water deficit. The crop and rangeland monitoring layer of the KE-ENSO Explorer, on the same CHIRPS v3 precipitation backbone as the rainfall and SPEI layers.

- **Temporal Coverage:** 2003 to present (open-ended)
- **Spatial Resolution:** kenya — 0.1 deg (~11 km, FEWS NET WRSI grid)

14.7.2 Data Producers & Custodians

- **Alliance of Bioversity International and CIAT** (Peter Steward)
- **USGS FEWS NET** - https://earlywarning.usgs.gov/fews

14.7.3 Scientific Caveats & Guidance

[!WARNING] Cropland and rangeland use different water-demand parameterisations and growing-season definitions (FEWS NET region codes / end-of-season dekads), so the two products are not interchangeable. The {crop} x {season} cells map to FEWS NET East Africa region codes as follows (FEWS NET zip-product instructions, Table 1, 2025-02-03): rangeland OND = e1 (Sep-Jan), rangeland MAM = e2 (Feb-Jul), cropland OND = et (maize, Oct-Feb), cropland MAM = ee (maize, Mar-Nov). Not every season x year combination of the template exists: the most recent season is only published once FEWS NET releases its

end-of-season product. PROVENANCE: objects published between 2026-08-18 and 2026-09-20 had the cropland and rangeland labels swapped (e1/e2 served as cropland, ek/et as rangeland); every object was replaced in place on 2026-09-20 and copies fetched before that date should be discarded.

14.8 `kenya-flood-exposure-intersect`: Kenya Flood x Exposure, Per-Sub-County Tables

Category: Satellite | **Publishing State:** `authored` | **License:** Open Data

14.8.1 Summary

Pre-computed flood x exposure statistics for Kenya's 290 IEBC sub-counties (admin 2), as three small Parquet tables the KE-ENSO Explorer reads directly (DuckDB-WASM) instead of intersecting heavy rasters and vectors in the browser. Two hazard products - Copernicus GFM observed flood (per season x year, 2018 to 2025) and JRC modelled return-period flood hazard - are each crossed with population (WorldPop constrained 100 m grid, levelled to the official KNBS county totals), classified roads and the electricity transmission grid (length exposed) and health and school facilities (count exposed); a third table holds the per-sub-county denominators. Tables carry no geometry: join on adm2_pcode to the boundary record kenya-admin-codab.

- **Temporal Coverage:** 2018 to 2025
- **Spatial Resolution:** kenya — IEBC sub-counties (290 admin-2 units)

14.8.2 Data Producers & Custodians

- **Alliance of Bioversity International and CIAT** (Peter Steward)

14.8.3 Scientific Caveats & Guidance

[!WARNING] Derived aggregate statistics; no input geometry is redistributed. Please attribute the inputs: Copernicus GFM and JRC Global Flood Maps (Copernicus / European Commission), WorldPop constrained population (CC-BY-4.0), OpenStreetMap roads and HOTOSM facilities ((c) OpenStreetMap contributors, ODbL), KPLC grid (CC0-1.0, energydata.info), IEBC COD-AB boundaries (OCHA, CC-BY-3.0-IGO). In the GFM table, observed_pct is the SAR-observed share of the sub-county in that season: low values mean sparse observation, not absence of flooding. Road and grid kilometres are cell-resolution approximations (in-cell line length within flooded cells). Every table also carries adm1_pcode, adm1_name and adm2_name label columns. POPULATION

METHOD (issue #28), stated in full. Headcounts are NOT raw gridded pixel sums. Every published sub-county population is built as three explicit factors, each published as its own column:

[!WARNING] pop_adm2 = pop_adm2_grid x pop_scale_census x pop_growth_county

[!WARNING] (1) pop_adm2_grid - the 100 m gridded pixel sum, which supplies the SHARE of the county's people

[!WARNING] living in that sub-county, and nothing else.

[!WARNING] (2) pop_scale_census = census county total / gridded county total - replaces the LEVEL of the

[!WARNING] gridded surface with the enumerated KNBS 2019 census, county by county. The gridded surfaces

[!WARNING] run about 17 per cent above the census (WorldPop constrained ~55.2 M, GRID3 ~55.9 M, census

[!WARNING] 47,564,296), so this factor is ~0.855 nationally but spans 0.324-1.394 across individual

[!WARNING] counties (measured on the published grid, 2026-09-17): WorldPop exceeds the census by up to

[!WARNING] threefold in some counties and falls below it in others. Any aggregate therefore moves by a

[!WARNING] weighted mix of county factors, not by the national one - flood-exposed population sits mostly

[!WARNING] in low-factor counties, so the GFM exposed total fell about 35 per cent where the national

[!WARNING] denominator fell 14 per cent. No time component.

[!WARNING] (3) pop_growth_county = target county total / census county total - the county's proportional

[!WARNING] CHANGE OVER TIME, from the KNBS Vol XVI projections (record

[!WARNING] kenya-population-knbs-projections). Exactly 1.0 when the denominator is the census itself.

[!WARNING] County is the only level where any of this is defensible: COD-AB adm2 is 290 IEBC constituencies while the census reports 345 KNBS sub-counties, and only 183 names match (record kenya-population-knbs-census); KNBS does not project below county at all. FIXED-SHARE ASSUMPTION. Each sub-county's share of its county is held constant at the gridded 2020 distribution, so every sub-county in a county moves by exactly that county's percentage change. Sub-county-specific growth is NOT represented: a fast-urbanising sub-county and a depopulating one in the same county grow at the same rate here. Growth does vary between counties - the 2025 factors span 1.03 to 1.15 - but not within one. YEAR MATCHING (the published default from 2026-09-17). The GFM table records OBSERVED floods and carries a year, so a single fixed denominator is wrong in one direction or the other: the 2019 census undercounts a 2024 flood by roughly 9 per cent, and a 2025 projection overcounts a 2018 one by a similar margin. Each GFM row is therefore levelled against ITS OWN year's county population, and pop_year records which year was used. Years the KNBS projections cover (2020 onward) use them; 2018 and 2019 fall back to the enumerated census, because nothing official projects backwards from the 2020 base and inventing a figure would be worse than stating the fallback. Consequence for consumers: pop_source VARIES BY ROW in the GFM table - "knbs-census-2019" for 2018-2019 and "knbs-projection-" thereafter - so do not read one value and assume it applies to the file. The JRC table is a return-period hazard with no event year, and the totals table is a static denominator, so both take a single stated reference year (pop_year, same on every row). TWO DEFINITIONS OF THE GROWTH FACTOR, recorded in pop_method:

[!WARNING] county-growth-from- - pop_growth_county = projection(year) / projection(base year,

[!WARNING] 2020). The enumerated census remains the level and KNBS supplies only the shape of change;

[!WARNING] national 2025 lands near 51.96 M because the ~2.6 per cent step KNBS puts between census night

[!WARNING] (August 2019) and its own 2020 base is deliberately not imported.

[!WARNING] county-level - pop_growth_county = projection(year) / census(2019), so the county total IS the

[!WARNING] published official projection and national 2025 is 53,330,964 - the figure a Kenyan

[!WARNING] counterpart would quote - at the cost of superseding the census anchor with the KNBS base-level

[!WARNING] revision.

[!WARNING] pop_source names the population vintage (knbs-census-2019, or knbs-projection-); pop_grid_source names the surface supplying the share. Nothing is discarded: the raw gridded figures remain in pop_exposed_grid and pop_total_grid. pop_pct is identical under every option - the factors cancel.

14.9 `kenya-population-knbs-census`: KNBS 2019 Census Population Counts by County and Sub-county (Kenya)

Category: Demographics | **Publishing State:** `authored` | **License:** Open Data

14.9.1 Summary

Enumerated population from the 2019 Kenya Population and Housing Census (KNBS), as admin-level tables: national, the 47 counties keyed on IEBC COD-AB adm1_pcode, the 345 KNBS sub-county rows in the Bureau's own sub-county universe, and a county x 5-year age group x sex table. The official denominator for anything the KE-ENSO Explorer presents to a Kenyan audience: the census total is 47,564,296, where the two gridded surfaces the Explorer also carries (kenya-population-worldpop 55.2 M, kenya-population-grid3 55.9 M) sit about 17 per cent higher.

- **Temporal Coverage:** 2019
- **Spatial Resolution:** kenya — counties (47 admin-1 units); a sub-county table is included in the KNBS universe

14.9.2 Data Producers & Custodians

- **Alliance of Bioversity International and CIAT** (Peter Steward)
- **Kenya National Bureau of Statistics (KNBS)** - https://www.knbs.or.ke/

14.9.3 Scientific Caveats & Guidance

[!WARNING] LICENCE EVIDENCE. Verified 2026-09-18 against the HDX API: the dataset "Kenya: Population Per County and sub county from Census Report", published by KNBS's OWN HDX organisation (Kenya National Bureau of Statistics), carries license_id other-pd-nr, "Public Domain / No restrictions (CC0)", and its resource is the identical workbook this record ingests ("Distribution of Population by Sex, Number of Households, Land Area, Population Density and Sub County.xlsx"). That is an explicit act of dedication by the rights holder over this content, so CC0-1.0 is asserted here rather than a LicenseRef. Note that KNBS census VOLUMES carry a general "All rights reserved" notice; it does not govern this table, which KNBS itself released without restriction. Attribution

is given as a courtesy though CC0 does not require it. The supporting evidence gathered earlier still stands: Evidence for the licence, recorded because knbs.or.ke itself carries no terms page (the terms-and-conditions URL 404s and the footer asserts "All Rights Reserved"), which is misleading: (1) KNBS operates an Open License Agreement covering its Open Data Platform, granting a worldwide, royalty-free, non-exclusive licence to use, copy, modify, publish, adapt, distribute and create derivative works for commercial and non-commercial purposes, subject to attribution. (2) KNBS's own account on the Humanitarian Data Exchange published the SAME sub-county table used here - "Distribution of Population by Sex, Number of Households, Land Area, Population Density and Sub County" - under Public Domain / No restrictions (https://data.humdata.org/dataset/kenya-population-per-county-from-census-report-2019). (3) The same census figures are redistributed by UNFPA under CC-BY-3.0-IGO (HDX cod-ps-ken). Required attribution, carried by any product derived from this record: "This product was adapted from the Statistical Office of Kenya's information, which is licensed under the KNBS Open License Agreement."

[!WARNING] ADMIN LEVELS DIFFER BY LEVEL. County joins one-for-one with IEBC COD-AB (47 units, adm1_pcode). Sub-county does NOT: COD-AB adm2 is 290 IEBC constituencies while the census reports 345 KNBS sub-county rows (a different universe - constituency splits such as Embakasi East/West/Central against sub-county Embakasi, plus 12 forest and national-park units such as Aberdare Forest and Mt Elgon Forest). Normalised in-county name matching resolves 183 of the 290 COD-AB units, so the adm2 table is published in the KNBS universe, name-keyed, with adm2_pcode_codab filled only where a name matches exactly one COD-AB unit inside the same county and codab_match recording how each row resolved. Use county totals plus a gridded within-county share for any adm2 denominator; do not join this table to COD-AB sub-counties on name. Male + Female (47,562,772) falls 1,524 short of the published Total because KNBS suppresses the intersex count below national level; the residual is carried as pop_intersex. The age-and-sex table comes from HDX cod-ps-ken (UNFPA, CC-BY-3.0-IGO), which redistributes the same census (Volume III) with adm1 p-codes attached. DO NOT SWITCH THE SOURCE TO THE HDX COPY of the sub-county table, despite its cleaner licence status: that copy is defective. It carries 346 rows to this source's 345 because the Tharaka-Nithi COUNTY total (393,177) appears as if it were a sub-county, inflating the national figure to 47,957,473 against the published 47,564,296. This ingest uses the knbs.or.ke workbook, which reconciles exactly. (It also spells Kutulo as "Kotulo"; same value, harmless.)

14.10 `kenya-population-knbs-projections`: KNBS Official Population Projections by County, 2020-2045 (Kenya)

Category: Demographics | **Publishing State:** `authored` | **License:** Open Data

14.10.1 Summary

Kenya's official cohort-component population projections, from the 2019 KPHC Analytical Report on Population Projections (Volume XVI, KNBS 2022), tabulated from the report's Appendix 5 as county x year x 5-year age group x sex, keyed on IEBC COD-AB adm1_pcode. National trajectory 48.82 M (2020), 53.33 M (2025), 57.81 M (2030), 62.16 M (2035), 66.31 M (2040), 70.18 M (2045). Lets forward-framed flood and drought exposure use a population denominator for the year in question instead of a 2020 surface read as current.

- **Temporal Coverage:** 2020 to 2045
- **Spatial Resolution:** kenya — counties (47 admin-1 units)

14.10.2 Data Producers & Custodians

- **Alliance of Bioversity International and CIAT** (Peter Steward)
- **Kenya National Bureau of Statistics (KNBS)** - https://www.knbs.or.ke/

14.10.3 Scientific Caveats & Guidance

> [!WARNING] MIXED-STATUS RECORD - READ BEFORE REUSE (decision 2026-09-18). The source documents are KNBS publications, linked here and not hosted by the Atlas. Three DERIVED parquet tables were published to digital-atlas BEFORE the licence was correctly established; they reproduce KNBS figures, so they are NOT open data. They are documented here deliberately: leaving a public object undocumented is worse than describing its true status. They must not be treated as open data, and this record must NOT be submitted to cdh-catalog, until written permission exists from KNBS (info@knbs.or.ke / directorgeneral@knbs.or.ke). Permission request is deferred as of 2026-09-18, so this state persists by decision, not by oversight. HOW THE ATLAS USES THIS WITHOUT REDISTRIBUTING IT. The exposure chain applies projections as a dimensionless growth ratio on a census anchor, never as absolute published values: R/observational/_population_helpers.R, POP_METHOD=county-growth (the default at 7_zonal_exposure.R:138 and 7b_relevel_exposure_pop.R:64) computes projection_county(year) / projection_county(2020) and multiplies the CC0 census count by it, so no published artifact reproduces a KNBS projection. The alternative POP_METHOD=county-level reproduces the official county total exactly and must NOT be used for anything published. SOURCING - CORRECTED 2026-10-07. An earlier version of this note claimed the ratios are read from the Summary Report rather than Volume XVI. That was WRONG: it

recorded an intention from the 2026-09-18 decision that was never implemented. python/ingest_population_knbs_projections.py downloads exactly one PDF, Volume XVI (PDF_URL, one urlretrieve_checked call), and the Summary Report URL appears nowhere in python/ or R/. The published tables are their own evidence - they carry 2020-2035 annually plus 2040 and 2045, while the Summary Report carries only 2020-2025, 2030, 2035, 2040 and 2045, so 2026-2029 and 2031-2034 can only have come from Volume XVI. The ratios therefore derive from the rights-reserved document, and the ratio method alone is what keeps published artifacts clear of KNBS figures. Switching the source is not a one-line change: the live reference year is 2026, which the Summary Report does not contain, so it would mean constraining POP_REF_YEAR to covered years - moving the national denominator from 52,837,534 (2026) to 51,961,873 (2025). Tracked on issue #33.

[!WARNING] LICENCE - CORRECTED 2026-09-18, READ BEFORE REUSE OR REDISTRIBUTION. This record previously claimed the KNBS Open License Agreement. That was WRONG. Volume XVI was read directly on 2026-09-18 and its copyright page states: "(c) 2022 Kenya National Bureau of Statistics. All rights reserved. No part of this publication may be reproduced, distributed, or transmitted in any form or by any means … without the prior written permission of the Bureau." The KNBS Open License Agreement is scoped to data published on the KNBS Open Data Platform and does not reach a PDF published on knbs.or.ke, and a search of HDX on 2026-09-18 found NO licensed mirror of these projections anywhere - the UNFPA COD-PS product (cod-ps-ken, CC-BY-3.0-IGO) carries the 2019 census only and contains no projections, so there is no openly licensed substitute. CONSEQUENCE. Extracting these figures for analysis is ordinary research use. REDISTRIBUTING them, including as the derived parquet files this record points at, needs written permission from KNBS (info@knbs.or.ke / directorgeneral@knbs.or.ke, the addresses given in the report). Until that permission exists this record must NOT be submitted to cdh-catalog, and the published derivatives should be treated as pending clearance rather than as open data. Tracked in hazards_prototype issue 33. This constraint does NOT apply to the census counts - record kenya-population-knbs-census is CC0-1.0, released without restriction by KNBS's own HDX account.

[!WARNING] LICENCE EVIDENCE. Weaker footing than the census record, and the difference is worth stating. KNBS's Open License Agreement grants worldwide, royalty-free, non-exclusive rights to reuse, adapt, redistribute and create derivative works with attribution - but it is scoped to data published on the KNBS Open Data Platform, and Volume XVI is a PDF hosted on knbs.or.ke. Unlike the 2019 census tables, no licensed mirror of these projections exists anywhere (searched HDX, open.africa and the Kenya data portal, 2026-09-17): UNFPA's COD-PS covers the census, not Volume XVI. What is published here is a derived tabulation of figures printed in a public national statistical report, attributed, not a copy of the report. Confirm with KNBS before this record is contributed to cdh-catalog. Required attribution: "This product was adapted from the Statistical Office of Kenya's information, which is licensed under the

KNBS Open License Agreement."

[!WARNING] COUNTY ONLY. Volume XVI does not project below county - no sub-county name appears anywhere in the report - so this record stops at county. The KE-ENSO exposure tables (record kenya-flood-exposure-intersect) do carry sub-county figures for a projection year, and the way they get there is stated there in full: each sub-county's share of its county is held fixed at the gridded 2020 distribution and the whole county is moved by its own percentage change from this table (pop_growth_county), with the enumerated census supplying the level. That means no sub-county-specific growth is represented; growth varies between counties (2025 factors span 1.03 to 1.15) but is uniform within one. Anyone wanting genuine sub-county differentials needs a source that produces them - this one does not. YEAR COVERAGE IS UNEVEN. The report's annual panels cover 2020-2035; 2040 and 2045 appear only in the five-yearly milestone panel, so 2036-2039 and 2041-2044 are absent. TWO SOURCE DEFECTS, both handled and logged by the ingest rather than silently absorbed. (1) Busia 2030, age 80+, female reads 3,120 in the 2030-2035 panel against 5,799 in the two other panels that carry 2030; the printed row total (9,714) reconciles only with 5,799, so 5,799 is used. (2) Kiambu's 2026-2030 panel labels its open-ended top age group "80-84" where every other panel says "80+"; it is relabelled "80+", without which it would double-count against the overlapping panels. KNBS rounds each printed cell independently, so sums do not close exactly: age groups land within five people of the printed All Ages row, and the 47 counties sum to roughly 15 to 20 people below the national row in every year. The identical report is also hosted as …-Analytical-Report-on-Population-Projections-Vol-XVI.pdf (2024/05, 32.9 MB); its extracted text is identical to the 2023/09 upload used here.

14.11 `kenya-population-worldpop`: WorldPop Constrained Gridded Population 2020 (Kenya)

Category: Demographics | **Publishing State:** `authored` | **License:** Open Data

14.11.1 Summary

Gridded population count for Kenya, 2020, from WorldPop's constrained (top-down dasymetric) 100 m product, cropped to Kenya and written as a Cloud Optimized GeoTIFF. The primary population-exposure surface of the KE-ENSO Explorer flood and drought overlay; paired with the GRID3 / WOPR bottom-up surface (record kenya-population-grid3), which has the same grain but a different method.

- **Temporal Coverage:** 2020
- **Spatial Resolution:** kenya — ~100 m (3 arc-seconds)

14.11.2 Data Producers & Custodians

- **Alliance of Bioversity International and CIAT** (Peter Steward)
- **WorldPop, University of Southampton** - https://www.worldpop.org/

14.11.3 Scientific Caveats & Guidance

[!WARNING] Constrained = top-down redistribution of national totals onto built-settlement extents. Use for spatial distribution, not as a census-anchored count. Compare with the GRID3 / WOPR bottom-up surface (record kenya-population-grid3). DISCREPANCY AGAINST THE CENSUS: the UN-adjusted national total, ~55.2 M, is about 17 per cent above the enumerated KNBS 2019 census figure of 47,564,296 (record kenya-population-knbs-census). Any absolute headcount taken straight off this surface is therefore ~17 per cent above what a Kenyan user will compare it against. The KE-ENSO exposure tables handle this by taking the LEVEL from KNBS at county and only the within-county SHARE from this grid (record kenya-flood-exposure-intersect); shares and percentages are unaffected either way. An earlier version of this record quoted ~55.9 M, which is the GRID3 total, not this one.

14.12 `kenya-roads-osm`: Kenya Classified Road Network (OpenStreetMap)

Category: Spatial | **Publishing State:** `authored` | **License:** Open Data

14.12.1 Summary

Kenya's classified road network (motorway, trunk, primary, secondary and tertiary highways) extracted from the Geofabrik Kenya OpenStreetMap extract and delivered as GeoJSON. The road-access overlay of the KE-ENSO Explorer flood and drought exposure: the network that matters for access, not every track and footpath.

- **Temporal Coverage:** 2026-08
- **Spatial Resolution:** kenya

14.12.2 Data Producers & Custodians

- **Alliance of Bioversity International and CIAT** (Peter Steward)
- **OpenStreetMap contributors (Geofabrik extract)** - https://www.openstreetmap.org/copyright

14.12.3 Scientific Caveats & Guidance

[!WARNING] ODbL: attribution "(c) OpenStreetMap contributors" and share-alike apply to redistribution of the data. Classified highways only (motorway to tertiary); footways, tracks and unclassified roads are excluded to keep the overlay light. Snapshot of the Geofabrik extract as of August 2026.

14.13 `kenya-facilities-hotosm`: Kenya Health and Education Facilities (HOTOSM)

Category: Spatial | **Publishing State:** `authored` | **License:** Open Data

14.13.1 Summary

Health facilities and education facilities (schools) for Kenya from the Humanitarian OpenStreetMap Team exports on HDX (OpenStreetMap-derived), clipped to Kenya and delivered as two GeoJSON files. The critical-asset overlay of the KE-ENSO Explorer flood and drought exposure. HOTOSM is the open working substitute for the official registries (KMHFR health facilities, GIGA schools), whose APIs were not reachable when the layer was built.

- **Temporal Coverage:** 2026-08
- **Spatial Resolution:** kenya — Facility features (points, some building polygons)

14.13.2 Data Producers & Custodians

- **Alliance of Bioversity International and CIAT** (Peter Steward)
- **Humanitarian OpenStreetMap Team (via HDX)** - https://data.humdata.org/organization/hot

14.13.3 Scientific Caveats & Guidance

[!WARNING] ODbL: attribution "(c) OpenStreetMap contributors, via Humanitarian OpenStreetMap Team" and share-alike apply to redistribution of the data. Snapshot of the HDX export as of August 2026. OSM-derived completeness varies by region; this is not an authoritative facility registry.

14.14 `kenya-power-grid-kplc`: Kenya Electricity Transmission Grid (KPLC)

Category: Spatial | **Publishing State:** `authored` | **License:** Open Data

14.14.1 Summary

Kenya's electricity transmission network - 11, 33, 66, 132 and 220 kV lines - from the Kenya Power and Lighting Company dataset on energydata.info, merged into one line layer (141,205 features) with a voltage attribute and delivered as GeoJSON. The 132 and 220 kV lines are the national backbone (highest-value at-risk assets). The power-infrastructure overlay of the KE-ENSO Explorer flood and drought exposure. The source's "Unidentified Grid Network" layer is excluded; gridfinder's modelled medium-voltage distribution grid is a deferred complement, not included here.

- **Temporal Coverage:** 2026-08
- **Spatial Resolution:** kenya

14.14.2 Data Producers & Custodians

- **Alliance of Bioversity International and CIAT** (Peter Steward)
- **Kenya Power and Lighting Company (KPLC), via energydata.info** - https://energydata.info/dataset/kenya-kenya-electricity-network

14.14.3 Scientific Caveats & Guidance

> [!WARNING] Transmission lines only (11 / 33 / 66 / 132 / 220 kV), not the distribution or medium-voltage network. Snapshot of the energydata.info dataset as retrieved in August 2026. The S3 object is served as application/octet-stream; it parses as GeoJSON.

14.15 `kenya-admin-codab`: Kenya Administrative Boundaries, Counties and Sub-counties (IEBC COD-AB)

Category: Spatial | **Publishing State:** `authored` | **License:** Open Data

14.15.1 Summary

Official Kenya administrative boundaries - 47 counties (admin 1) and 290 sub-counties (admin 2) - from the OCHA Common Operational Dataset for Administrative Boundaries (COD-AB), source IEBC, with official p-codes, harmonised to EPSG:4326 and delivered as two GeoJSON files. The reporting-unit backbone for aggregating KE-ENSO Explorer flood and drought exposure (record kenya-flood-exposure-intersect joins on adm2_pcode). Preferred over GAUL for exposure because it carries p-codes and excludes disputed slivers.

- **Temporal Coverage:** 2019-10 to 2025-01
- **Spatial Resolution:** kenya — Admin 1 (47 counties) and admin 2 (290 sub-counties)

14.15.2 Data Producers & Custodians

- **Alliance of Bioversity International and CIAT** (Peter Steward)
- **OCHA / Independent Electoral and Boundaries Commission (IEBC), via HDX** - https://data.humdata.org/dataset/cod-ab-ken

14.15.3 Scientific Caveats & Guidance

> [!WARNING] Geometries were made valid and reprojected but not simplified or otherwise altered. Attribute columns follow the COD-AB convention (adm0/adm1/adm2 _pcode and _name, valid_on, area_sqkm).

14.16 `crop-vop-intld15`: MapSPAM 2020 Adaptation Atlas, Sub-Saharan Africa

Category: Agri | **Publishing State:** `authored` | **License:** Open Data

14.16.1 Summary

Spatially disaggregated crop statistics for Sub-Saharan Africa: 42 crops across six production systems, harmonised to FAOSTAT country totals, as physical area, harvested area, production, yield and value of production. A regional adaptation of IFPRI's MapSPAM 2020 built for the Africa Agriculture Adaptation Atlas, supplementing the global release with national and sub-national sources. This is the crop-production basis of the Atlas exposure chain: every crop value-of- production layer, and through them the hazard-exposure product, is distributed across these production shares.

- **Temporal Coverage:** 2020
- **Spatial Resolution:** sub-saharan-africa

14.16.2 Data Producers & Custodians

- **Alliance of Bioversity International and CIAT** - https://alliancebioversityciat.org/
- **International Food Policy Research Institute (IFPRI)** - https://www.ifpri.org/
- **Alliance of Bioversity International and CIAT** (Peter Steward)

14.16.3 Scientific Caveats & Guidance

[!WARNING] NOT THE SAME DATASET AS THE HUB'S `spam2020` RECORD, DESPITE THE NAME. That record describes IFPRI's GLOBAL MapSPAM 2020 v2r2, licensed CC-BY-SA-4.0 and published as zarr. This one is the Atlas's Sub-Saharan Africa adaptation, licensed CC-BY-4.0 and published as GeoTIFF. Different scope, different licence, different format, different DOI. Do not substitute one for the other, and note that the share-alike term on the global release does NOT reach the Atlas exposure chain, which consumes this product. THREE PROCESSING LEVELS, AND THEY ARE NOT INTERCHANGEABLE. `raw` holds the released CSVs, one file per variable and technology code (P_TA, P_TH, P_TI and so on). `analysis-ready` holds rasterised GeoTIFFs. `atlas-harmonized` holds the Atlas value-of-production layers derived from them. The VoP chain reads `analysis-ready/variable=prod_t` and writes into `atlas-harmonized`. FILE NAMING DIFFERS BETWEEN LEVELS. The irrigated system is `irrigated` under `analysis-ready` and `irr` under `atlas-harmonized`. Anything templating a path across both levels must special-case it; the token values below are given per level for that reason. CURRENCY BASIS IS A KNOWN TRAP. Several value-of-production variables coexist - vop_intld15, vop_usd2015, vop_nominal-usd15, vop_nominal-usd21 - and they are NOT interchangeable. A past mismatch between the crop and livestock currency bases produced roughly a sevenfold error in cattle exposure. Constant international dollars (vop_intld15) is the Atlas standard; check the livestock counterpart is on the same basis before combining. Tracked as hazards_prototype issue 30. THREE OBJECTS ARE PUBLISHED UNDER A MALFORMED KEY. Directly beneath `region=ssa/` there are three parquet objects whose path reads `processing=atlas-harmonizedvariable=<name>.parquet`, missing the separator between the two hive fields: crop-livestock_all, vop_intld15-2021 and vop_nominal-usd-2021. They are not reachable by a hive-partitioned scan and should be republished or removed rather than left as a trap for the next reader.

14.17 `livestock-vop`: Gridded Livestock of the World (GLW4) Value of Production (Adaptation Atlas SSA)

Category: Agri | **Publishing State:** `authored` | **License:** Open Data

14.17.1 Summary

Spatial density distribution and economic value of production for major livestock species (cattle, sheep, goats, camels, pigs, chickens) across Sub-Saharan Africa. Based on FAO Gridded Livestock of the World v4 (GLW4 2020) downscaled to 1 km / 5-minute resolution, harmonised to FAOSTAT country production values and national livestock populations, expressed in constant 2015 international dollars (vop_intld15).

- **Temporal Coverage:** 2020
- **Spatial Resolution:** sub-saharan-africa, kenya — 1 km (0.008333 deg)

14.17.2 Data Producers & Custodians

- **Food and Agriculture Organization of the United Nations (FAO)** - https://www.fao.org/livestock-systems/global-distributions/en/
- **Alliance of Bioversity International and CIAT** (Peter Steward)

14.17.3 Scientific Caveats & Guidance

> [!WARNING] Attribution: "Data provided by FAO Gridded Livestock of the World (GLW4) and compiled for the African Agriculture Adaptation Atlas (AAAA)." CURRENCY BASIS MUST REMAIN CONSTANT INTERNATIONAL DOLLARS (vop_intld15). Several value-of-production metrics coexist (nominal USD, constant 2015 international dollars). To avoid severe distortions when pairing livestock with MapSPAM crop VoP, both portfolios must be evaluated on the identical constant 2015 international dollar basis. Past unit mismatches produced ~7x errors in cattle exposure. DENSITY VS PIXEL COUNTS. GLW4-2020 layers represent per-km2 animal density (D-DA), not raw counts per grid cell. Spatial aggregation across county administrative polygons must weight by grid-cell land area (excluding open waterbodies). LIVESTOCK VULNERABILITY LAGS. Unlike annual field crops which fail within a single dry dekad, pastoral herds exhibit biological inertia: mortality, distress sales, and conception deficits lag rainfall anomalies by 6 to 12 months.

14.18 `knbs-napr`: KNBS National Agricultural Production Reports (Crops, Livestock & Products)

Category: Agri | **Publishing State:** `authored` | **License:** Open Data

14.18.1 Summary

Audited subnational agricultural statistics from the Kenya National Bureau of Statistics (KNBS) National Agricultural Production Reports (NAPR 2023/24 and 2024/25 editions). Contains county-level annual area harvested (ha), production (t), and gross farm-gate value (KES) across 22 major food and cash commodities (maize, beans, sorghum, tea, coffee, potatoes, pyrethrum, wheat, dairy milk, beef), cross-verified via dual-engine deterministic parsing (PyMuPDF) and additivity gates.

- **Temporal Coverage:** 2019 to 2024
- **Spatial Resolution:** kenya

14.18.2 Data Producers & Custodians

- **Kenya National Bureau of Statistics (KNBS)** - https://www.knbs.or.ke/
- **Alliance of Bioversity International and CIAT** (Peter Steward)

14.18.3 Scientific Caveats & Guidance

> [!WARNING] Attribution: "Data extracted deterministically from the Kenya National Bureau of Statistics (KNBS) National Agricultural Production Reports (2023/24, 2024/25), Republic of Kenya." PROJECT RULE D1 COMPLIANCE: ZERO LLM INFERENCE. Every number served from this dataset is parsed directly from published government PDF tables by deterministic code (_sources/napr_extract.py) and passed through an automated machine additivity gate. No language model reads, types, or imputes any numerical value. BLANK IS NOT ZERO. KNBS data is compiled from administrative expert-estimates. Missing county-years represent unobserved reporting gaps, never zero production. Records with missing data are preserved as null and never imputed as false zeroes. CROSS-EDITION REBASE (2025 EDITION PRIORITY). Where reporting years overlap between the 2023/24 and 2024/25 editions, the latest edition is prioritised because it incorporates KNBS retrospective prior-year revisions. ADMINISTRATIVE ESTIMATES VS CENSUS SURVEYS. NAPR figures represent annual ministerial and county directorate of agriculture administrative estimates, not full-enumeration agricultural censuses.

14.19 `harveststat-crop-production`: HarvestStat Africa - Harmonized Subnational Crop Statistics

Category: Agri | **Publishing State:** `authored` | **License:** Open Data

14.19.1 Summary

Cleaned and harmonised subnational crop production, harvested area and yield for Sub-Saharan Africa, assembled from FEWS NET and FAO sources: 574,204 records, 33 countries, 94 crop types, primarily 1980-2022. A federated record - the Adaptation Atlas cites this dataset but does not host or redistribute it.

- **Temporal Coverage:** None
- **Spatial Resolution:** None

14.19.2 Data Producers & Custodians

- **NASA Harvest / University of Maryland** - https://github.com/HarvestStat/HarvestStat-Africa
- **Famine Early Warning Systems Network (FEWS NET)** - https://fews.net/
- **Alliance of Bioversity International and CIAT** (Peter Steward)

14.19.3 Scientific Caveats & Guidance

> [!WARNING] LICENCE CONFIRMED 2026-09-18. The Dryad landing page does not print a licence, which is why this record was initially held as a draft. The Dryad API settles it: the dataset record returns license = https://spdx.org/licenses/CC0-1.0.html, at version 12, last modified 2026-07-17. CC0 requires no attribution; the citation below is given anyway. Quantities are metric tonnes, areas hectares, yields metric tonnes per hectare. Coverage is jagged across countries, crops and years - check per-country completeness before treating any national series as continuous.

14.20 `kenya-market-prices-fews`: Kenya Retail and Wholesale Market Prices (FEWS NET Data Warehouse)

Category: Markets | **Publishing State:** `authored` | **License:** Open Data

14.20.1 Summary

Staple and livestock market prices for Kenya from the FEWS NET Data Warehouse market price facts: 42 counties by market by product, 12 products, retail and wholesale, in Kenyan shillings, 2000 to 2026. Pulled through the FDW API. The price basis behind the Kenya ENSO Explorer's market panel and behind the pastoral terms of trade shown there, which the Atlas derives rather than reads.

- **Temporal Coverage:** 2000 to present (open-ended)
- **Spatial Resolution:** kenya

14.20.2 Data Producers & Custodians

- **Famine Early Warning Systems Network (FEWS NET), USAID** - https://fews.net/
- **Alliance of Bioversity International and CIAT** (Peter Steward)

14.20.3 Scientific Caveats & Guidance

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14.21 `kenya-market-prices-kamis`: Kenya Live-Animal Market Prices by Grade and Sex (MoALD KAMIS)

Category: Markets | **Publishing State:** `draft` | **License:** Open Data

14.21.1 Summary

Market-day livestock price quotes for Kenya from the Kilimo AgriMarkets Market Information System (KAMIS) of the Ministry of Agriculture and Livestock Development (MoALD): cattle, sheep, goats and camels priced per head at named livestock markets in 45 counties, each quote carrying a breed (classification), a quality grade (1 to 3), the animal's sex and the volume on offer, from May 2021 to the present. Harvested from the portal's search export. The only verified open Kenyan source that prices camels and sheep and that records grade and sex, which the FEWS NET series behind the Kenya ENSO Explorer does not; it extends the explorer's pastoral price panel from goats and cattle to all four pastoral species at market-day frequency.

- **Temporal Coverage:** 2021-05-24 to present (open-ended)
- **Spatial Resolution:** kenya

14.21.2 Data Producers & Custodians

- **Ministry of Agriculture and Livestock Development (MoALD), Kenya** - https://kamis.kilimo.go.ke/
- **Alliance of Bioversity International and CIAT** (Peter Steward)

14.21.3 Scientific Caveats & Guidance

> [!WARNING] NO PUBLISHED LICENCE. The KAMIS portal (kamis.kilimo.go.ke) footer asserts "Copyright (c) KAMIS" and gives the Ministry's contact (info@kilimo.go.ke, +254-20-2718870, Cathedral Road, Nairobi); it carries no licence statement and no terms of use, and the About page credits the National Agricultural and Rural Inclusive Growth Project (NARIGP) for the system. A copyright line without a licence is a reservation, not a grant. The Atlas reads the quotes for analysis and displays derived monthly medians; the market-day quotes are not republished to S3 or to the Climate Data Hub and no bulk download is offered until MoALD confirms terms of reuse. HOW THE PORTAL PAGES, AND WHY EARLIER COUNTS WERE WRONG. The HTML results table pages by an offset in the URL path (/site/market_search/), not by the page= query parameter, so one wide request returns only the first per_page rows. The Excel export honours per_page and returns a whole window in one file. The harvester pulls one export per species-year and checks the row count against the portal's own pagination total; on 2026-10-23 that was 116,634 rows across the four species. WHOLESALE IS THE LIVESTOCK PRICE. 99.8 percent of

live-animal quotes sit in the portal's Wholesale column, which is the per-head price at the livestock market; the Retail column is essentially empty for animals. The explorer therefore serves these as wholesale prices, distinct from the FEWS NET retail goat and cattle series it already shows. THE 2021-05-24 BULK BACKFILL. The portal's history starts on 2021-05-24, and that single date carries 11,079 quotes, about one tenth of the whole archive. It is the upload date of a backfill, not a market day. Those rows are kept in the market-day file with the flag launch_day_bulk and are excluded from the monthly medians. KAMIS cannot reach the 2011 or 2017 drought analogues; the FEWS NET series remains the long-run record. DIRTY ROWS ARE FLAGGED, NOT DROPPED. A county literally named test (345 quotes), 3,203 exact duplicate submissions, 67 non-positive prices and 591 order-of-magnitude outliers against the same market-day median are all present in the market-day file with a qc_flag, and absent from the monthly medians. County spellings are canonicalised to the 47-county key (Tana-River to Tana River, Homa-bay to Homa Bay, Muranga to Murang'a); the portal spelling is kept in county_raw. COVERAGE IS UNEVEN AND THINNING. Reporting peaked in 2022 (about 40,000 quotes) and fell to a few thousand a year by 2025; camel quotes stop at 2025-11-24. Absence of a market-month is a gap in reporting, not a zero price. Grade and breed are enumerator judgements with no published grading standard, so grade comparisons across markets are indicative only. NO COORDINATES. The portal names the market and county only; the file is plain parquet, not GeoParquet. OBFUSCATED DELIVERY, BY CHOICE. As a courtesy to the Ministry while reuse terms are sought, the market-day file is published to the explorer only as an AES-256-GCM container (market_prices_kamis.parquet.enc) that the explorer decrypts in the browser; the monthly medians it displays are derived aggregates. This stops casual download and crawling. It is not access control: a static page must hold whatever decrypts it, so a determined reader can recover the bytes. The plain file is not committed to the repository and no bulk download is offered.

14.22 `kenya-food-insecurity-ipc`: Kenya Acute Food Insecurity Phase Classification (IPC / KFSSG)

Category: Markets | **Publishing State:** `authored` | **License:** Open Data

14.22.1 Summary

IPC Acute Food Insecurity phase classifications and population-in-phase tables for Kenya, produced twice a year by the Kenya Food Security Steering Group after the long-rains and short-rains assessments. A federated record - the Adaptation Atlas cites this analysis but does not host it.

- **Temporal Coverage:** None
- **Spatial Resolution:** None

14.22.2 Data Producers & Custodians

- **Integrated Food Security Phase Classification (IPC) Global Partnership** - https://www.ipcinfo.org/
- **Kenya Food Security Steering Group (KFSSG)**
- **Alliance of Bioversity International and CIAT** (Peter Steward)

14.22.3 Scientific Caveats & Guidance

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14.23 `kenya-county-gender-datasheets`: Kenya County Gender Data Sheets, 2025

Category: Demographics | **Publishing State:** `authored` | **License:** Open Data

14.23.1 Summary

Gender and social-inclusion indicators for all 47 Kenyan counties, published as one data sheet per county in October 2025. Indicators cover health and fertility, education, economic participation, political representation, disability and attitudes to intimate-partner violence, drawn largely from the 2022 Kenya Demographic and Health Survey, the 2019 census and

the 2022 general election. A federated record - the Adaptation Atlas reads these sheets but does not host or redistribute them.

- **Temporal Coverage:** None
- **Spatial Resolution:** kenya

14.23.2 Data Producers & Custodians

- **Kenya National Bureau of Statistics (KNBS)** - https://www.knbs.or.ke/
- **State Department for Gender and Affirmative Action, Ministry of Gender, Culture and Children Services**
- **Council of Governors**
- **UN Women (Women Count)** - https://data.unwomen.org/women-count
- **SDGs Kenya Forum**
- **Alliance of Bioversity International and CIAT** (Peter Steward)

14.23.3 Scientific Caveats & Guidance

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table. Treating the whole table as “2022” is wrong for some indicators. BLANK IS NOT ZERO. The extraction serves a series only where at least 40 of 47 counties agree with the modal national value; non-matching county rows are dropped individually. So a served series can be silently missing up to seven counties, and a missing county-series is a parse or gate drop - NEVER an observed zero. Only 10 of 35 series reach all 47 counties; the worst is households with an elderly member at 40/47. Per-county row counts run 22-35. COVERAGE IS ASYMMETRIC WITHIN AN INDICATOR. One code can have 47 counties for male and 46 for female. Do not compute a sex ratio or a gender gap without checking both sides are present for that county. CHART-ONLY INDICATORS ARE ABSENT BY METHOD. Antenatal-visit and vaccination charts and internet usage are dropped by the gate. Their absence reflects the extraction, not the source. LABELS ARE NOT DEFINITIONS. The canonical label is the longest string seen across the 47 sheets and several remain truncated mid-sentence. Do not lift them into variable descriptions. COLUMN-ORDER FRAGILITY. Values are read Kenya-then-County from the printed header. A future edition that flips that header would swap every value silently; the coverage gate catches it only if the national column also becomes inconsistent.

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15 Controlled Vocabulary & Scientific Glossary

16 Controlled Vocabulary & Scientific Glossary

This chapter provides standardized canonical definitions, abbreviations, mathematical notations, and institutional citations for all **65 domain terms** across the Kenya ENSO Explorer platform. It establishes the controlled terminology required for interoperable subnational climate risk analytics.

16.1 Administrative & Geography

16.1.1 Arid and Semi-Arid Lands (ASAL)

The 23 designated arid and semi-arid counties of Kenya comprising over 80% of the national landmass, characterized by low and erratic rainfall, fragile rangeland ecosystems, and predominantly pastoralist livelihoods.

Mathematical Formulation / Metric:

$$\text{County Count} = 23 \text{ designated ASAL counties}$$

Canonical Institutional / Literature Reference: *Kenya Ministry of Devolution and ASALs.*

Active in Platform Sections: `0.0`, `1.0`, `1.1`, `4.3`, `4.4`

16.1.2 Common Operational Datasets - Administrative Boundaries (COD-AB)

The authoritative United Nations humanitarian GIS administrative boundary layers (ADM0 National, ADM1 County, ADM2 Sub-county) endorsed by UN OCHA and the Kenya Survey Department.

Mathematical Formulation / Metric:

$$Vectorpolygonalboundarylayerswithstandardized P-codes$$

Canonical Institutional / Literature Reference: *UN OCHA Humanitarian Data Exchange (HDX).*

Active in Platform Sections: `1.0`, `1.1`, `3.4`, `6.1`

16.1.3 Humanitarian OpenStreetMap Team Facilities (HOT-OSM)

Community-mapped geospatial vector points of health clinics, hospitals, and educational institutions in Kenya, validated for humanitarian disaster exposure modeling.

Mathematical Formulation / Metric:

$$Geospatialpointfeaturelayer$$

Canonical Institutional / Literature Reference: *Humanitarian OpenStreetMap Team (HOT) Kenya Export.*

Active in Platform Sections: `3.5`, `4.5`, `6.1`

16.1.4 Kenya Power and Lighting Company Grid (KPLC)

National electrical transmission line infrastructure vector dataset used in the platform to evaluate power grid asset vulnerability to flood inundation.

Mathematical Formulation / Metric:

$$Vectorlinenetwork(kVtransmissionlines)$$

Canonical Institutional / Literature Reference: *Kenya Power and Lighting Company GIS Repository.*

Active in Platform Sections: `4.5`, `6.1`

16.2 Climatology & Seasons

16.2.1 Long Rains (MAM)

The primary agricultural rainfall season across western, central, and northern Kenya, spanning the months of March, April, and May, driven by the northward migration of the Intertropical Convergence Zone (ITCZ).

Mathematical Formulation / Metric:

$$R_{\text{MAM}} = \sum_{m \in \{\text{Mar, Apr, May}\}} R_m$$

Canonical Institutional / Literature Reference: *Kenya Meteorological Department (KMD) National Seasonal Calendar.*

Active in Platform Sections: `0.0`, `1.0`, `2.1`, `3.1`, `3.1B`, `4.1`

16.2.2 Pastoral Boreal Summer Window (JJAS / JAS)

The dry season across eastern and southern pastoral rangelands (June–September) and the primary rainfall season in the western highlands and Lake Victoria basin.

Mathematical Formulation / Metric:

$$R_{\text{JJAS}} = \sum_{m \in \{\text{Jun, Jul, Aug, Sep}\}} R_m$$

Canonical Institutional / Literature Reference: *Kenya Meteorological Department (KMD) Regional Climatology.*

Active in Platform Sections: `1.0`, `3.1`, `4.3`

16.2.3 Short Rains (OND)

The secondary agricultural and pastoral rainfall season spanning October, November, and December, exhibiting the highest teleconnection sensitivity and predictive skill linked to Pacific ENSO and Indian Ocean Dipole dynamics.

Mathematical Formulation / Metric:

$$R_{\text{OND}} = \sum_{m \in \{\text{Oct, Nov, Dec}\}} R_m$$

Canonical Institutional / Literature Reference: *Kenya Meteorological Department (KMD) National Seasonal Calendar.*

Active in Platform Sections: `0.0`, `1.0`, `2.1`, `3.1`, `3.1B`, `4.1`

16.2.4 WMO 1991–2020 Standard Normal Baseline (WMO Normal)

The canonical 30-year climatological reference period defined by the World Meteorological Organization for assessing contemporary climate anomalies and evaluating extreme frequencies.

Mathematical Formulation / Metric:

$$Calendar period 1991-01-01 through 2020-12-31 (30 complete historical years)$$

Canonical Institutional / Literature Reference: *WMO-No. 1203 (2017). WMO Guidelines on the Calculation of Climate Normals.*

Active in Platform Sections: `0.0`, `2.1`, `3.1`, `3.1B`, `3.2`, `6.1`

16.2.5 Warm Dry Season Window (DJF)

The warm dry interval spanning December, January, and February, marked by crop maturation and harvesting in agricultural counties and grazing pressure in pastoral zones.

Mathematical Formulation / Metric:

$$R_{\mathrm{DJF}} = R_{\mathrm{Dec}} + R_{\mathrm{Jan}} + R_{\mathrm{Feb}}$$

Canonical Institutional / Literature Reference: *Kenya Meteorological Department (KMD).*

Active in Platform Sections: `1.0`, `3.1`

16.3 Demography & Equity

16.3.1 Gender Equality and Social Inclusion (GESI)

A developmental framework and empirical indicator suite examining how gender headship, intra-household decision-making, land tenure documentation, and labor division influence differential vulnerability to climate extremes.

Mathematical Formulation / Metric:

$$Multidimensional indicator suite benchmarking county vs national medians$$

Canonical Institutional / Literature Reference: *KNBS County Gender Data Sheets (2025).*

Active in Platform Sections: `1.3`, `6.1`

16.4 Economic & Valuation

16.4.1 Gross Value of Production (VOP)

The aggregate physical output of agricultural crops and livestock commodities valued at constant standardized gate prices, providing an uninflated metric of total agricultural capital at risk from climate extremes.

Mathematical Formulation / Metric:

$$\text{VOP}_c = \sum_{k=1}^{K} Q_{c,k} \times P_k^{\text{constant}}$$

Canonical Institutional / Literature Reference: *FAOSTAT Valuation Methodology / Decision D62 Beta Review Governance.*

Active in Platform Sections: `1.2`, `4.5`

16.5 Food Security & Vulnerability

16.5.1 Integrated Food Security Phase Classification (IPC)

A multi-partner standardized scale categorizing food insecurity severity into five distinct phases: Phase 1 (Minimal), Phase 2 (Stressed), Phase 3 (Crisis), Phase 4 (Emergency), and Phase 5 (Catastrophe/Famine).

Mathematical Formulation / Metric:

$$\text{Phase } \in \{1, 2, 3, 4, 5\}$$

Canonical Institutional / Literature Reference: *IPC Global Partners (2021). Integrated Food Security Phase Classification Technical Manual Version 3.1.*

Active in Platform Sections: `4.6`, `6.1`

16.5.2 Reduced Coping Strategy Index (rCSI)

A standardized proxy measure of household food stress based on a weighted 5-question survey assessing behavioral adaptations taken by households experiencing food shortages over the past 7 days.

Mathematical Formulation / Metric:

$$\text{rCSI} = \sum_{i=1}^{5} w_i \times f_i, \quad w \in \{1, 1, 1, 2, 3\}$$

Canonical Institutional / Literature Reference: *Maxwell, D., et al. (2008). The Coping Strategies Index Field Methods Manual. CARE / WFP.*

Active in Platform Sections: `4.6`, `6.1`

16.6 Institutional & Governance

16.6.1 Agriculture and Food Authority (AFA)

The state regulator overseeing development, regulation, and promotion of scheduled agricultural crops in Kenya (including sugar, tea, coffee, nuts, and cereals).

Mathematical Formulation / Metric:

$$State regulatory authority$$

Canonical Institutional / Literature Reference: *Agriculture and Food Authority Act (2013).*

Active in Platform Sections: `4.1`, `6.1`

16.6.2 Anticipatory Action (AA)

Acting ahead of predicted hazardous climate events to prevent or reduce humanitarian impacts before disaster strikes, triggered by probabilistic meteorological forecasts.

Mathematical Formulation / Metric:

$$\text{Trigger: } P(\text{Extreme}) \geq \tau_{\text{action}}$$

Canonical Institutional / Literature Reference: *UN OCHA / Red Cross Climate Centre Anticipatory Action Framework.*

Active in Platform Sections: `0.0`, `2.2`

16.6.3 Comprehensive Data Hub Schema (CDH Schema)

The institutional metadata and data publishing specification (v0.3.0) governing dataset descriptions, citation formatting, temporal-spatial schemas, and open data licensing across the Adaptation Atlas.

Mathematical Formulation / Metric:

$$Metadata contract specification v0.3.0$$

Canonical Institutional / Literature Reference: *Africa Agriculture Adaptation Atlas CDH Technical Specification.*

Active in Platform Sections: `6.1`

16.6.4 Disaster Risk Reduction (DRR)

Policy objective aimed at preventing new and reducing existing disaster risk and managing residual risk, strengthening resilience.

Mathematical Formulation / Metric:

$$SendaiFrameworkforDisasterRiskReduction2015-2030.$$

Canonical Institutional / Literature Reference: *UNDRR Sendai Framework.*

Active in Platform Sections: `0.0`, `2.2`, `4.6`

16.6.5 Famine Early Warning Systems Network (FEWS NET)

A leading provider of early warning and analysis on food insecurity created by USAID, providing data on markets, trade, and agro-climatology across East Africa.

Mathematical Formulation / Metric:

$$Globalearlywarningnetwork$$

Canonical Institutional / Literature Reference: *USAID FEWS NET.*

Active in Platform Sections: `4.1`, `4.2B`, `4.4`, `4.4B`, `6.1`

16.6.6 Food and Agriculture Organization (FAO / FAOSTAT)

The specialized agency of the United Nations leading international efforts to defeat hunger, maintain global agricultural statistical databases (FAOSTAT), and establish standardized commodity price concepts.

Mathematical Formulation / Metric:

$$UNSpecializedStatisticalAgency$$

Canonical Institutional / Literature Reference: *FAO Statistical Databases (FAOSTAT).*

Active in Platform Sections: `1.2`, `4.1`, `6.1`

16.6.7 Kenya Agricultural and Livestock Research Organization (KALRO)

The principal agricultural research institution in Kenya developing crop varieties, rangeland management protocols, and agro-weather advisories.

Mathematical Formulation / Metric:

$$Nationalagriculturalresearchauthority$$

Canonical Institutional / Literature Reference: *KALRO Act (2013).*

Active in Platform Sections: `4.1`, `6.1`

16.6.8 Kenya Food Security Steering Group (KFSSG)

The multi-agency technical coordination body led by NDMA and national line ministries that conducts biannual post-harvest food security assessments (Short Rains Assessment - SRA; Long Rains Assessment - LRA).

Mathematical Formulation / Metric:

$$Biannualconsensusassessments(SRA/LRA)$$

Canonical Institutional / Literature Reference: *National Drought Management Authority (NDMA).*

Active in Platform Sections: `0.0`, `4.6`, `6.1`

16.6.9 Kenya Meteorological Department (KMD)

The canonical national hydrometeorological service of Kenya responsible for national weather forecasting, seasonal climate outlooks, marine telemetry, and severe weather early warnings (Decision D40 canonical institutional naming).

Mathematical Formulation / Metric:

$$Nationalhydrometauthority$$

Canonical Institutional / Literature Reference: *Ministry of Environment, Climate Change and Forestry.*

Active in Platform Sections: `0.0`, `2.1`, `2.2`, `2.3`, `6.1`

16.6.10 Kenya National Bureau of Statistics (KNBS)

The principal government agency responsible for collecting, analyzing, and disseminating official statistical information in Kenya, including the decennial Population and Housing Census and economic surveys.

Mathematical Formulation / Metric:

$$National statistical authority$$

Canonical Institutional / Literature Reference: *Statistics Act (2006).*

Active in Platform Sections: `1.0`, `1.1`, `1.2`, `1.3`, `4.1`, `6.1`

16.6.11 National Drought Management Authority (NDMA)

The statutory leadership agency in the Government of Kenya responsible for drought early warning, crisis coordination, resilience building, and operating the national early warning bulletin monitoring network.

Mathematical Formulation / Metric:

$$Monthly drought monitoring across 23 ASAL counties$$

Canonical Institutional / Literature Reference: *National Drought Management Authority Act (2016).*

Active in Platform Sections: `0.0`, `1.2`, `4.6`, `6.1`

16.6.12 Regional Centre for Mapping of Resources for Development (RCMRD)

An intergovernmental organization based in Nairobi, Kenya, providing geospatial and remote sensing support, disaster risk mapping, and Earth observation analytics across Eastern and Southern Africa.

Mathematical Formulation / Metric:

$$Intergovernmental Earth observation center$$

Canonical Institutional / Literature Reference: *RCMRD Institutional Charter.*

Active in Platform Sections: `3.5`, `6.1`

16.6.13 ReliefWeb Disaster Chronicle (ReliefWeb / GLIDE)

The UN OCHA humanitarian information portal tracking disaster situations and compiling emergency response dispatches indexed by Global Identifier Numbers (GLIDE).

Mathematical Formulation / Metric:

$$GLIDEUniqueDisasterIdentifierstandard$$

Canonical Institutional / Literature Reference: *UN OCHA ReliefWeb Service.*

Active in Platform Sections: `4.6`, `6.1`

16.7 Markets & Early Warning

16.7.1 Cross-Border Food Trade (XBT Trade)

The formal and informal monitored cross-border trade of staple cereal grains and livestock across East African border corridors (e.g. Busia, Malaba, Moyale, Namanga, Mandera) tracked quarterly by FEWS NET.

Mathematical Formulation / Metric:

$$V_{\text{XBT}} = \sum \text{Tonnes}_{\text{cereal}} \text{ (Import / Export)}$$

Canonical Institutional / Literature Reference: *FEWS NET East Africa Cross-Border Trade Bulletin.*

Active in Platform Sections: `4.4B`, `6.1`

16.7.2 Kenya Agricultural Market Information System (KAMIS)

The national market price surveillance platform operated by the Ministry of Agriculture and Livestock Development tracking daily wholesale and retail food commodity prices across Kenya's urban and rural markets.

Mathematical Formulation / Metric:

$$Daily/weeklymarketsurveillancetimeseries$$

Canonical Institutional / Literature Reference: *Ministry of Agriculture and Livestock Development (MoALD).*

Active in Platform Sections: `4.4`, `6.1`

16.7.3 Pastoral Terms of Trade (ToT)

The ratio of pastoral purchasing power, measured as the number of kilograms of local staple grain (dry maize) that a pastoral household can purchase through the sale of one medium-sized live goat (shoat).

Mathematical Formulation / Metric:

$$\mathrm{ToT}_m = \frac{P_{\mathrm{goat},m}}{P_{\mathrm{maize},m}} \quad (\mathrm{kg/goat})$$

Canonical Institutional / Literature Reference: *National Drought Management Authority (NDMA) Monthly Early Warning Bulletins.*

Active in Platform Sections: `4.4`, `6.1`

16.8 Ocean & Climate Drivers

16.8.1 Columbia IRI Prediction Plume (IRI Plume)

A multi-model ensemble of dynamical and statistical seasonal forecast models predicting Niño 3.4 SST anomalies up to 9 months ahead, curated monthly by the International Research Institute for Climate and Society (IRI).

Mathematical Formulation / Metric:

$$\bar{z}_{\mathrm{IRI}} = \frac{1}{M} \sum_{m=1}^{M} z_{\mathrm{model}_m}$$

Canonical Institutional / Literature Reference: *Columbia Climate School International Research Institute for Climate and Society.*

Active in Platform Sections: `2.2`

16.8.2 Dipole Mode Index (DMI)

The quantitative measure of the Indian Ocean Dipole, defined as the difference in sea surface temperature anomaly between the western tropical Indian Ocean (50°E–70°E, 10°S–10°N) and the southeastern tropical Indian Ocean (90°E–110°E, 10°S–0°).

Mathematical Formulation / Metric:

$$\mathrm{DMI} = \mathrm{SSTA}(50^\circ\text{–}70^\circ\mathrm{E}, 10^\circ\mathrm{S}\text{–}10^\circ\mathrm{N}) - \mathrm{SSTA}(90^\circ\text{–}110^\circ\mathrm{E}, 10^\circ\mathrm{S}\text{–}0^\circ)$$

Canonical Institutional / Literature Reference: *JAMSTEC Application Laboratory / NOAA Physical Sciences Laboratory.*

Active in Platform Sections: `0.0`, `2.1`, `3.2`, `3.3`, `4.1`

16.8.3 El Niño–Southern Oscillation (ENSO)

A coupled ocean-atmosphere phenomenon in the tropical Pacific characterized by irregular fluctuations between warm (El Niño), cool (La Niña), and neutral sea surface temperature anomalies, exerting global atmospheric teleconnections.

Mathematical Formulation / Metric:

$$\mathrm{SSTA}_{\text{Niño 3.4}} \geq +0.5^{\circ}\mathrm{C}\ \text{(El Niño)}, \leq -0.5^{\circ}\mathrm{C}\ \text{(La Niña)}$$

Canonical Institutional / Literature Reference: *Trenberth, K. E. (1997). The definition of El Niño. BAMS, 78(12), 2771-2778.*

Active in Platform Sections: `0.0`, `2.1`, `2.2`, `3.1`, `3.2`, `3.3`, `4.1`, `5.1`

16.8.4 Indian Ocean Dipole (IOD)

An ocean-atmosphere coupled climate mode in the equatorial Indian Ocean characterized by an anomalous sea surface temperature gradient between the western tropical Indian Ocean and the eastern tropical Indian Ocean.

Mathematical Formulation / Metric:

$$\mathrm{DMI} = \mathrm{SSTA}_{\text{Western IO}} - \mathrm{SSTA}_{\text{Eastern IO}}$$

Canonical Institutional / Literature Reference: *Saji, N. H., et al. (1999). A dipole mode in the tropical Indian Ocean. Nature, 401(6751), 360-363.*

Active in Platform Sections: `0.0`, `2.1`, `2.2`, `3.2`, `3.3`, `4.1`

16.8.5 Oceanic Niño Index (ONI)

NOAA's primary operational index for tracking ENSO, computed as the 3-month running mean of ERSST.v5 SST anomalies in the Niño 3.4 region (5°N–5°S, 120°–170°W) based on 30-year centered base periods.

Mathematical Formulation / Metric:

$$\mathrm{ONI} = \frac{1}{3}\sum_{m=1}^{3} \mathrm{SSTA}_{\text{Niño 3.4},m}$$

Canonical Institutional / Literature Reference: *NOAA Climate Prediction Center (CPC) Operational ENSO Criteria.*

Active in Platform Sections: `2.1`, `3.2`, `3.3`

16.8.6 Relative Oceanic Niño Index (RONI)

The Oceanic Niño Index (Niño 3.4 SST anomaly) minus the mean tropical ocean SST anomaly (20°S–20°N). Measures relative warmth against tropical background warming to identify true atmospheric convective forcing in a warming climate.

Mathematical Formulation / Metric:

$$\text{RONI} = \text{SSTA}_{\text{Niño 3.4}} - \text{SSTA}_{\text{Tropical (20°S–20°N)}}$$

Canonical Institutional / Literature Reference: *van Oldenborgh, G. J., et al. (2021). Relative SST indices for ENSO. Climate Dynamics.*

Active in Platform Sections: `0.0`, `2.1`, `3.2`, `3.3`, `4.1`

16.8.7 SINTEX-F Forecast Plume (SINTEX-F)

The Scale Interaction Experiment-Frontier coupled ocean-atmosphere general circulation model developed by JAMSTEC, used operationally for 9-month lead forecasts of the Indian Ocean Dipole and ENSO.

Mathematical Formulation / Metric:

$$Ensemble mean of 9 coupled dynamical simulation members$$

Canonical Institutional / Literature Reference: *Luo, J. J., et al. (2005). Extended-range ENSO and IOD predictions using SINTEX-F. J. Climate.*

Active in Platform Sections: `2.2`

16.8.8 Sea Surface Temperature Anomaly (SSTA)

The deviation of ocean surface temperature from the long-term climatological mean for a given location and calendar month, typically referenced to a 30-year WMO normal baseline.

Mathematical Formulation / Metric:

$$\text{SSTA}_{y,m} = \text{SST}_{y,m} - \bar{\text{SST}}_{m(1991\text{–}2020)}$$

Canonical Institutional / Literature Reference: *WMO Climatological Standard Normals Technical Document No. 1203.*

Active in Platform Sections: `2.1`, `3.3`

16.8.9 Western Pacific V-shape Pattern (Western-V / WV)

A recognized sea surface temperature and atmospheric circulation dipole in the Western North Pacific that modulates moisture flux into the Greater Horn of Africa, particularly relevant during boreal spring (MAM).

Mathematical Formulation / Metric:

$$\text{WV Index} = \text{Standardized } z\text{-score of WNP SST/Wind anomaly dipole}$$

Canonical Institutional / Literature Reference: *Funk, C., et al. (2023). Western Pacific driver mechanics modulating East African rainfall.*

Active in Platform Sections: `2.1`, `5.1`

16.9 Official Statistics

16.9.1 HarvestStat Subnational Crop Statistics (HarvestStat)

A harmonized, multi-decadal historical subnational crop production database developed by the FEWS NET Science Team and UC Santa Barbara, extending from 1982 through 2020 for Kenyan counties.

Mathematical Formulation / Metric:

$$Harmonized historical time series with quality flags$$

Canonical Institutional / Literature Reference: *FEWS NET Science Team & Climate Hazards Center.*

Active in Platform Sections: `4.2B`, `6.1`

16.9.2 National Agriculture Production Report (KNBS NAPR)

The official national agricultural survey published by the Kenya National Bureau of Statistics providing subnational, county-level production statistics across 47 counties for crops, livestock headcounts, and animal products.

Mathematical Formulation / Metric:

$$Official state statistical release (2024/2025 editions)$$

Canonical Institutional / Literature Reference: *Kenya National Bureau of Statistics (KNBS).*

Active in Platform Sections: `1.1`, `4.1`, `6.1`

16.10 Remote Sensing & Hazards

16.10.1 Climate Hazards Center InfraRed Precipitation with Station Data (CHIRPS v3)

A quasi-global high-resolution precipitation dataset blending cold cloud duration (CCD) satellite infrared imagery with calibrated in-situ station observations, providing gridded rainfall at 0.05° spatial resolution from 1981 to present.

Mathematical Formulation / Metric:

$$R_{\text{CHIRPS}} = R_{\text{IR}} \times \frac{S_{\text{gauged}}}{S_{\text{climatology}}}$$

Canonical Institutional / Literature Reference: *Funk, C., et al. (2015). The climate hazards infrared precipitation with stations. Scientific Data, 2, 150066.*

Active in Platform Sections: `0.0`, `1.0`, `2.1`, `3.1`, `3.4`, `6.1`

16.10.2 Cold Cloud Duration (CCD)

The duration of time a satellite thermal infrared pixel registers temperatures below a convective threshold (typically -40°C), used by CHIRPS as a physical proxy for convective precipitation.

Mathematical Formulation / Metric:

$$\text{CCD} = \sum_{t} \mathbb{I}(T_b(t) < 233.15\,\text{K}) \times \Delta t$$

Canonical Institutional / Literature Reference: *Milford, J. R., & Dugdale, G. (1990). Estimation of rainfall using cold cloud duration. Int. J. Remote Sens.*

Active in Platform Sections: `3.1`, `3.4`

16.10.3 Copernicus Global Flood Monitoring (Copernicus GFM)

An operational Earth observation service providing near-real-time satellite Synthetic Aperture Radar (SAR) flood inundation extents derived from Sentinel-1 constellation radar acquisitions at 20m resolution, independent of cloud cover.

Mathematical Formulation / Metric:

$$\text{SAR Backscatter Threshold: } \sigma_0 < \tau_{\text{water}} \text{ (dB)}$$

Canonical Institutional / Literature Reference: *European Commission Joint Research Centre (JRC) Copernicus GFM Service.*

Active in Platform Sections: `1.0`, `3.5`, `4.5`, `6.1`

16.10.4 Cropland Land Cover Mask (Cropland Mask)

A 10m spatial resolution binary landcover mask derived from ESA WorldCover Sentinel-1/Sentinel-2 classification identifying cultivated agricultural parcels.

Mathematical Formulation / Metric:

$$M_{\text{crop}}(x, y) \in \{0, 1\}$$

Canonical Institutional / Literature Reference: *Zanaga, D., et al. (2021). ESA WorldCover 10m 2020 v100.*

Active in Platform Sections: `1.2`, `4.5`

16.10.5 Gridded Livestock of the World (GLW / Livestock Density)

Global high-resolution modeled spatial distribution grids of cattle, goats, sheep, and camels disaggregated to sub-kilometer pixels using random forest environmental regressions.

Mathematical Formulation / Metric:

$$D_{\text{animal}}(x, y) \text{ (heads/km}^2\text{)}$$

Canonical Institutional / Literature Reference: *Gilbert, M., et al. (2018). Global distribution data for cattle, buffaloes, horses, sheep, goats, pigs, chickens and ducks in 2010. Scientific Data.*

Active in Platform Sections: `1.2`, `4.5`

16.10.6 JRC Flood Return Period Hazard Layers (JRC Flood RP)

Hydrodynamic inundation depth maps modeled by the European Commission Joint Research Centre for statistical return intervals (1-in-10, 1-in-20, 1-in-50, 1-in-100 years) based on calibrated river discharge distributions.

Mathematical Formulation / Metric:

$$T_{\text{RP}} = \frac{1}{P(\text{Exceedance in any single year})}$$

Canonical Institutional / Literature Reference: *Dottori, F., et al. (2016). Development and evaluation of a framework for global flood hazard mapping. Advances in Water Resources.*

Active in Platform Sections: `3.5`, `4.5`, `6.1`

16.10.7 Normalized Difference Vegetation Index (MODIS NDVI)

A satellite remote sensing index quantifying photosynthetic biomass and vegetation greenness by computing the normalized ratio of surface reflectance in near-infrared (NIR) and visible red (RED) spectrums.

Mathematical Formulation / Metric:

$$\mathrm{NDVI} = \frac{\rho_{\mathrm{NIR}} - \rho_{\mathrm{RED}}}{\rho_{\mathrm{NIR}} + \rho_{\mathrm{RED}}}$$

Canonical Institutional / Literature Reference: *Rouse, J. W., et al. (1974). Monitoring the vernal advancement of retrogradation of natural vegetation. NASA/GSFC.*

Active in Platform Sections: `4.3`, `6.1`

16.10.8 Standardized Precipitation Index (SPI)

A widely used drought index based on the probability of precipitation for any time scale, fitted to a gamma distribution and transformed to a normal distribution with mean zero and standard deviation one.

Mathematical Formulation / Metric:

$$\mathrm{SPI} = \Phi^{-1}(G(P))$$

Canonical Institutional / Literature Reference: *McKee, T. B., et al. (1993). The relationship of drought frequency and duration to time scales. AMS.*

Active in Platform Sections: `3.1`, `6.1`

16.10.9 Standardized Precipitation-Evapotranspiration Index (SPEI-3)

A multiscalar drought index that combines precipitation and atmospheric evaporative demand (potential evapotranspiration) accumulated over a 3-month rolling window, fitted to a log-logistic probability distribution.

Mathematical Formulation / Metric:

$$D_i = P_i - \mathrm{PET}_i, \quad \mathrm{SPEI} = \Phi^{-1}(F(D))$$

Canonical Institutional / Literature Reference: *Vicente-Serrano, S. M., et al. (2010). A multiscalar drought index sensitive to global warming: SPEI. J. Climate.*

Active in Platform Sections: `3.1`, `6.1`

16.10.10 Synthetic Aperture Radar (SAR)

An active microwave imaging sensor that illuminates the Earth's surface with radar pulses, capable of penetrating cloud cover, dense smoke, and nighttime darkness to detect smooth open water surfaces via specular reflection.

Mathematical Formulation / Metric:

$$\sigma^0 \text{ (radar backscatter coefficient in decibels)}$$

Canonical Institutional / Literature Reference: *Sentinel-1 Technical Guide, European Space Agency (ESA).*

Active in Platform Sections: `1.0`, `3.5`, `4.5`

16.10.11 Water Requirement Satisfaction Index (WRSI)

A geospatial agro-climatological crop water balance model calculating the ratio of actual seasonal evapotranspiration to crop water requirements, used by FEWS NET to identify crop and pasture moisture stress.

Mathematical Formulation / Metric:

$$\text{WRSI} = \frac{\sum_{t=1}^{N} \text{ET}_a}{\sum_{t=1}^{N} \text{ET}_c} \times 100\%$$

Canonical Institutional / Literature Reference: *Senay, G. B., & Verdin, J. (2003). Characterizing drought conditions in the Greater Horn of Africa using WRSI. Int. J. Remote Sens.*

Active in Platform Sections: `4.1`, `6.1`

16.11 Statistical Methods

16.11.1 Bivariate Joint Probability Distribution (Joint Probability)

The empirical probability of two climate variables concurrently occupying designated states (e.g., El Niño in the Pacific co-occurring with positive Indian Ocean Dipole).

Mathematical Formulation / Metric:

$$P(A \cap B) = \frac{n(A \cap B)}{N_{\text{total}}}$$

Canonical Institutional / Literature Reference: *Wilks, D. S. (2011). Statistical Methods in the Atmospheric Sciences.*

Active in Platform Sections: `3.2`, `4.1`

16.11.2 Empirical Confidence Interval (95% CI)

The range of statistical estimates within which the true population parameter is expected to fall with 95% probability, calculated via non-parametric bootstrapping.

Mathematical Formulation / Metric:

$$\mathrm{CI}_{95\%} = [\hat{Q}_{\mathrm{boot}}(0.025), \hat{Q}_{\mathrm{boot}}(0.975)]$$

Canonical Institutional / Literature Reference: *Efron, B., & Tibshirani, R. J. (1994). An Introduction to the Bootstrap.*

Active in Platform Sections: `2.2`, `3.3`

16.11.3 Hyndman-Fan Type 7 Quantile Estimator (Hyndman-Fan Type 7)

The standard continuous empirical quantile estimation algorithm used in R, Python, and DuckDB where sample fractiles are linearly interpolated between adjacent sorted values.

Mathematical Formulation / Metric:

$$\hat{Q}(p) = X_{(\lfloor h \rfloor)} + (h - \lfloor h \rfloor)(X_{(\lfloor h \rfloor + 1)} - X_{(\lfloor h \rfloor)}), \quad h = (N-1)p + 1$$

Canonical Institutional / Literature Reference: *Hyndman, R. J., & Fan, Y. (1996). Sample quantiles in statistical packages. The American Statistician.*

Active in Platform Sections: `2.1`, `3.1`

16.11.4 Independent Seasonal Standardization (Seasonal z-score)

A standardization protocol where OND and MAM seasonal rainfall are transformed into z-scores independently against their own distinct historical seasonal means and standard deviations, ensuring fair multi-season sequence classification.

Mathematical Formulation / Metric:

$$z_{s,y} = \frac{R_{s,y} - \mu_{s(1991\text{–}2020)}}{\sigma_{s(1991\text{–}2020)}}, \quad s \in \{\mathrm{OND}, \mathrm{MAM}\}$$

Canonical Institutional / Literature Reference: *Decision D53 Technical Architecture Specification.*

Active in Platform Sections: `3.1B`

16.11.5 Leave-One-Out Cross-Validation (LOOCV)

A deterministic cross-validation procedure where each year in the historical hindcast record is iteratively omitted from the training distribution and used solely as the out-of-sample evaluation target, preventing artificial hindcast overconfidence.

Mathematical Formulation / Metric:

$$\text{Skill}_{\text{LOOCV}} = \frac{1}{N}\sum_{y=1}^{N} \mathcal{S}(\hat{P}_{(-y)}, O_y)$$

Canonical Institutional / Literature Reference: *Wilks, D. S. (2011). Statistical Methods in the Atmospheric Sciences. Academic Press.*

Active in Platform Sections: `0.0`, `2.2`, `6.1`

16.11.6 Modal Hit Rate (Modal Hit Rate)

The percentage of historical verification seasons where the forecast's highest-probability (modal) category exactly matched the tercile category that actually occurred.

Mathematical Formulation / Metric:

$$\text{HR} = \frac{1}{N}\sum_{i=1}^{N} \mathbb{I}\left(\arg\max_k p_{i,k} == \arg\max_k o_{i,k}\right) \times 100\%$$

Canonical Institutional / Literature Reference: *Jolliffe, I. T., & Stephenson, D. B. (2012). Forecast Verification: A Practitioner's Guide.*

Active in Platform Sections: `0.0`, `2.2`

16.11.7 Non-Parametric Empirical Quantiles (Empirical Quantiles)

Fractile estimators calculated directly from the sorted empirical cumulative distribution function without assuming a theoretical parametric distribution (such as Gaussian normal). Mandatory for East African rainfall due to positive skewness.

Mathematical Formulation / Metric:

$$\hat{Q}(p) = (1-\gamma)X_{(j)} + \gamma X_{(j+1)}, \quad j = \lfloor pN + (1-p)/3 \rfloor$$

Canonical Institutional / Literature Reference: *Hyndman, R. J., & Fan, Y. (1996). Sample quantiles in statistical packages. The American Statistician.*

Active in Platform Sections: `2.1`, `3.1`, `3.2`

16.11.8 Olympic Mean Baseline (Olympic Mean)

A robust central tendency statistic calculated by dropping the highest and lowest extreme values from a multi-year sample (e.g. 5 or 10 years) and averaging the remaining values, minimizing distortion from black-swan shocks.

Mathematical Formulation / Metric:

$$\bar{X}_{\text{Olympic}} = \frac{1}{N-2} \sum_{i=2}^{N-1} X_{(i)}$$

Canonical Institutional / Literature Reference: *WMO Climatological Practices Guide.*

Active in Platform Sections: `4.1`, `4.2B`

16.11.9 Ordinary Least Squares Linear Regression (OLS Regression)

A parametric linear model estimating the slope and intercept relating continuous oceanic driver z-scores to county rainfall accumulation in mm.

Mathematical Formulation / Metric:

$$\hat{Y} = \beta_0 + \beta_1 X + \epsilon, \quad \min_{\beta} \sum (Y_i - \hat{Y}_i)^2$$

Canonical Institutional / Literature Reference: *Weisberg, S. (2005). Applied Linear Regression.*

Active in Platform Sections: `3.3`

16.11.10 Rainfall Positive Skewness (Positive Skewness)

The asymmetric right-tailed distribution characteristic of East African seasonal precipitation, where the majority of seasons experience near- or below-average totals and a minority experience extreme episodic deluge.

Mathematical Formulation / Metric:

$$\gamma_1 = \frac{\frac{1}{N}\sum_{i=1}^{N}(R_i - \bar{R})^3}{\left(\frac{1}{N}\sum_{i=1}^{N}(R_i - \bar{R})^2\right)^{3/2}} > 0$$

Canonical Institutional / Literature Reference: *Hastings, D., et al. (2020). Skewness in African rainfall extremes. J. Hydrometeorology.*

Active in Platform Sections: `3.1`, `6.1`

16.11.11 Ranked Probability Skill Score (RPSS)

A categorical probabilistic verification metric that measures the relative accuracy of a multicategory probabilistic forecast (such as terciles) compared to a climatological reference forecast.

Mathematical Formulation / Metric:

$$\text{RPS} = \sum_{m=1}^{K} \left(\sum_{k=1}^{m} p_k - \sum_{k=1}^{m} o_k \right)^2, \quad \text{RPSS} = 1 - \frac{\text{RPS}_{\text{forecast}}}{\text{RPS}_{\text{climatology}}}$$

Canonical Institutional / Literature Reference: *Epstein, E. S. (1969). A scoring system for probability forecasts of ranked categories. J. Appl. Meteor.*

Active in Platform Sections: `0.0`, `2.2`, `6.1`

16.11.12 Standard Normal Homogeneity Test (SNHT)

A statistical test applied to climatological time series to detect abrupt shifts or changepoints caused by station relocations, instrument recalibrations, or landcover changes.

Mathematical Formulation / Metric:

$$T_k = k\bar{z}_1^2 + (n-k)\bar{z}_2^2$$

Canonical Institutional / Literature Reference: *Alexandersson, H. (1986). A homogeneity test applied to precipitation data. J. Climatology.*

Active in Platform Sections: `3.1`

16.11.13 Subcounty Spatial Apportionment (Spatial Apportionment)

The mathematical disaggregation of county-level aggregate production and livestock values across subcounties using high-resolution spatial auxiliary weights.

Mathematical Formulation / Metric:

$$V_{sc} = V_c \times w_{sc}, \quad \sum_{sc \in c} w_{sc} = 1.0$$

Canonical Institutional / Literature Reference: *Decision D62 Technical Valuation Specification.*

Active in Platform Sections: `1.2`

16.11.14 Teleconnection Gradient (Pearson r)

The bivariate linear correlation coefficient quantifying the strength and sign of teleconnection between ocean sea surface temperature anomalies and local county precipitation.

Mathematical Formulation / Metric:

$$r = \frac{\sum (z_x - \bar{z}_x)(z_y - \bar{z}_y)}{\sqrt{\sum (z_x - \bar{z}_x)^2 \sum (z_y - \bar{z}_y)^2}}$$

Canonical Institutional / Literature Reference: *Pearson, K. (1895). Mathematical contributions to the theory of evolution.*

Active in Platform Sections: `3.3`
